SHORT-TIME EXISTENCE, UNIQUENESS AND SMOOTH DEPENDENCE FOR NONLINEAR PARABOLIC EQUATIONS OF ARBITRARY EVEN ORDER ON VECTOR BUNDLES

[AUTHORS]

Abstract. We prove that a fully nonlinear evolution equation $\partial_t u = F(t, u, \nabla u, \dots, \nabla^{2p} u)$ for sections of a vector bundle over a closed Riemannian manifold has, for every smooth initial datum at which the equation is parabolic, a unique smooth solution for a short time, and that this solution depends smoothly on the initial datum. Parabolicity is expressed by a pointwise algebraic *Legendre condition* on the linearization. It covers the quasilinear equations of product type treated by Mantegazza and Martinazzi, polyharmonic flows of every order, the Ricci–DeTurck flow and parabolic Monge–Ampère equations.

The proof is self-contained modulo standard facts about Sobolev spaces on closed manifolds, and it is organized with a machine formalization in mind:

- all function spaces have integer order;
- no trace, interpolation or fractional spaces occur;
- linear theory is needed only for operators with smooth coefficients;
- regularity of solutions follows from the implicit function theorem via Angenent's parameter trick, not from tame estimates.

Status. This is a working draft intended to serve as the mathematical blueprint of a Lean formalization. Results labelled "Fact" are standard and are imported rather than proved here; Section 8 lists them together with the dependency structure of the argument. Places where the draft still needs checking are marked [like this].

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1. Introduction

Let (M, g) be a closed Riemannian manifold and $V \to M$ a vector bundle equipped with a fibre metric and a compatible connection. We study evolution equations

$$\partial_t u = F\big(t, u, \nabla u, \dots, \nabla^{2p} u\big), \qquad u(\cdot, 0) = u_0, \tag{1.1}$$

for time-dependent sections u of V, where $p \geq 1$ and F is smooth. Equations of this form occur throughout geometric analysis:

- order two ($p = 1$): the harmonic map heat flow and the Ricci–DeTurck flow;
- order four: Willmore-type flows, Calabi-type flows, the Q-curvature flow and the biharmonic map heat flow;
- arbitrary even order: polyharmonic flows.

Almost every study of such a flow starts from the same statement: for a smooth initial datum at which the equation is parabolic there is a unique smooth solution for a short time, and it depends continuously on the datum. That statement is almost always justified by an appeal to "standard parabolic theory".

For second-order equations and systems there is a large literature behind this appeal, for instance [14, 11, 31, 1, 23, 4]. It is usually set in Euclidean domains, in Hölder spaces or in an abstract semigroup framework, and the statement on a closed manifold has to be extracted from it. For higher-order equations the situation is thinner.

- Polden [28] proposed a proof based on linear theory in parabolic Sobolev spaces and on the inverse function theorem.

- Sharples [29] pointed out a gap in the convergence of the frozen-coefficient linear problems used there.
- Mantegazza and Martinazzi [25] closed the gap for scalar quasilinear equations whose leading coefficient has the product form $E_1 \otimes \cdots \otimes E_p$, with Polden's linear theory used as a black box. They also remarked that systems can be treated in the same way.

The purpose of this paper is to give one self-contained statement and proof that covers all of the following at once:

- equations of any even order $2p$;
- sections of an arbitrary vector bundle with metric and compatible connection;
- fully nonlinear dependence on all derivatives up to order $2p$;
- parabolicity assumed only along the initial datum;
- uniqueness, and smooth dependence on the initial datum.

1.1. **The main result.** In the informal statement below, $P_{u_0} = \partial F/\partial(\nabla^{2p} u)$ denotes the coefficient of the top-order derivative in the linearization of F at the initial datum.

Theorem 1.1 (informal version of Theorems A–C)**.** *Suppose that at every point of M the coefficient P_{u_0} satisfies the Legendre condition of Definition 2.6. Then:*

(1) equation (1.1) has a smooth solution on $M \times [0,T]$ for some $T > 0$;
(2) two smooth solutions along which the equation stays parabolic coincide;
(3) the solution depends smoothly on u_0.

The Legendre condition asks that the quadratic form $\xi \mapsto (-1)^{p+1} \sum_{I,J} \langle \xi_I, P^{IJ} \xi_J \rangle$ be positive definite on V-valued p-tensors ξ. It is the higher-order analogue of the Legendre condition of the calculus of variations.

- It holds for $-(\nabla^*\nabla)^p$, and for every leading coefficient of product form, in particular for the class treated in [25] (Lemma 2.8).
- It is stable under the usual rearrangements of covariant derivatives (Remark 2.9).
- It is strictly stronger than Petrovskii's strong parabolicity in general, but equivalent to it for scalar second-order equations and in several other cases (Remark 2.10).

1.2. **Strategy.** We keep the central idea of [25]. The nonlinear operator is a C^1 map between parabolic Sobolev spaces; we linearize it at a smooth *approximate* solution and apply the inverse function theorem. Around this core we make five changes. Each one either widens the scope or removes a technical ingredient.

(1) *Approximate solutions from the formal power series.* We use the Taylor polynomial in t of the formal power series solution (Lemma 5.1), rather than the solution of an auxiliary linear problem. This construction works verbatim for fully nonlinear equations.
(2) *No trace spaces.* The unknown is sought in the space $\mathcal{P}^m_0$ of functions vanishing at $t = 0$, as a perturbation of the approximate solution. The error of the approximate solution is cut off near $t = 0$ by a smooth cutoff (Lemma 5.2). The sharp trace space of the parabolic Sobolev space never appears.
(3) *Linear theory proved, not quoted.* We obtain energy estimates at every integer level by differentiating the equation. Differentiation moves the problem from V to $T^*M \otimes V$, which is why working with general bundles is natural even for scalar equations. The model operator $-(\nabla^*\nabla)^p$ is inverted by eigenfunction expansion, and general operators are reached by the method of continuity (Section 4). Negative-order norms are defined by duality, and all orders are integers.
(4) *Regularity by Angenent's trick.* Pulling a solution back by the flows of horizontal lifts of vector fields produces a family of equations depending smoothly on a parameter.

The implicit function theorem, applied at a smooth point, shows that the solution depends smoothly on the parameter, and this dependence is exactly spatial regularity (Section 5.3; compare [2]). Consequently neither tame estimates nor linear theory with non-smooth coefficients is ever needed.

(5) *Uniqueness by linearization.* The difference of two solutions solves a linear equation with smooth coefficients, and uniqueness follows from the basic energy estimate (Section 5.4).

1.3. **A proof designed to be formalized.** This paper is written as the blueprint of a formalization in Lean 4, on top of Mathlib [26] and an existing library of differential geometry [9]. Several of the choices above were made for that reason.

- Every function space has integer order and is defined through covariant derivatives and time derivatives.
- There are no interpolation spaces, trace theorems, Fourier multipliers or pseudodifferential operators.
- Because the data are smooth, we may work in spaces of very high order and use only crude embeddings.
- The linear theory is only ever applied to operators with smooth coefficients.
- The ellipticity hypothesis is a pointwise algebraic condition that can be checked directly.
- The final theorems are stated purely in terms of smooth sections, covariant derivatives and time derivatives. The Sobolev machinery is internal to the proofs.

1.4. **What is not covered.** The following are deliberately outside the scope of this paper.

- Manifolds with boundary.
- Noncompact manifolds.
- Rough initial data and optimal regularity.
- Strongly parabolic operators that admit no Legendre representation (Remark 2.10).
- Flows that are degenerate because of a gauge symmetry: the Ricci flow, the mean curvature and Willmore flows of immersions, the Yang–Mills flow. Our theorems apply to these after gauge fixing (for instance the Ricci–DeTurck flow, Section 6.3). The step that recovers the original flow from the gauge-fixed one is not treated here; see Section 7.

1.5. **Organization.**

- Section 2: the setting, the Legendre condition and the main theorems.
- Section 3: the function spaces.
- Section 4: linear theory.
- Section 5: the nonlinear theorems.
- Section 6: examples.
- Section 7: remarks on the scope.
- Section 8: the dependency structure and the imported facts.
- Appendix A: the abstract tools.

2. Setting and statements

2.1. **Bundles and covariant derivatives.** Throughout, (M, g) is a closed Riemannian manifold, that is, compact and without boundary, of dimension $n \geq 1$, with Riemannian measure μ.

A *Riemannian bundle* is a real vector bundle $W \to M$ with a fibre metric $\langle \cdot, \cdot \rangle$ and a metric connection ∇. The Levi-Civita connection of g and the connection of W induce metric connections on all bundles $\otimes^k T^*M \otimes W$. All of them are denoted by ∇.

Iterated derivatives. For $u \in \Gamma(W)$ we define $\nabla^k u \in \Gamma(\otimes^k T^*M \otimes W)$ inductively by $\nabla^0 u = u$ and

$$(\nabla^k u)(X_1, \dots, X_k) = \big(\nabla_{X_1}(\nabla^{k-1} u)\big)(X_2, \dots, X_k).$$

Thus the first slot records the derivative taken last, and $\nabla^k(\nabla^l u) = \nabla^{k+l} u$ as tensors.

Frames and multi-indices. In a local orthonormal frame (e_i) with dual coframe (e^i), we write $T_{i_1 \dots i_k} \in W_x$ for the components of a W-valued k-tensor. For a multi-index $I = (i_1, \dots, i_p)$ we write $e^I = e^{i_1} \otimes \cdots \otimes e^{i_p}$.

Permutations. A permutation $\sigma \in S_k$ acts on W-valued k-tensors by $(\sigma T)(X_1, \dots, X_k) = T(X_{\sigma(1)}, \dots, X_{\sigma(k)})$. This is a parallel bundle isometry.

Integration. We write $\langle u, v \rangle = \int_M \langle u, v \rangle \, d\mu$ and $\|u\|$ for the L^2 inner product and norm. The formal adjoint of $\nabla \colon \Gamma(W) \to \Gamma(T^*M \otimes W)$ is

$$\nabla^* \eta = -\sum_i (\nabla \eta)_{ii},$$

where the derivative slot is contracted with the first slot of η. By the divergence theorem, $\langle \nabla u, \eta \rangle = \langle u, \nabla^* \eta \rangle$. The connection Laplacian is $\nabla^* \nabla u = -\sum_i (\nabla^2 u)_{ii}$.

Time-dependent sections. A time-dependent section on $[0, T]$ is a section of the pullback of W to $M \times [0, T]$, and $C^\infty(M \times [0, T]; W)$ denotes the smooth ones, smooth up to $t = 0$ and $t = T$. The operator ∇ acts in the M-variables. Since g and the connection do not depend on t, ∂_t commutes with ∇ and with all the bundle operations above.

Differential operators. For Riemannian bundles W, W' and $r \in \mathbb{N}$, the space $\mathrm{Diff}^r(W, W')$ consists of the operators

$$(Au) = \sum_{k=0}^{r} A_k(\nabla^k u), \qquad A_k \in C^\infty\big(M \times [0, T]; \mathrm{Hom}(\otimes^k T^*M \otimes W, W')\big).$$

Their coefficients may depend on t. By the Leibniz rule, $\mathrm{Diff}^r \circ \mathrm{Diff}^s \subseteq \mathrm{Diff}^{r+s}$.

Lemma 2.1 (commuting covariant derivatives)**.** *Let W be a Riemannian bundle, $r \geq 2$ and $\sigma \in S_r$. There is an operator $Q_\sigma \in \mathrm{Diff}^{r-2}(W, \otimes^r T^*M \otimes W)$ such that*

$$\sigma(\nabla^r u) = \nabla^r u + Q_\sigma u \qquad \text{for all } u \in \Gamma(W).$$

Its coefficients are universal polynomial expressions in the curvatures of g and of W and in their covariant derivatives of order at most $r - 2$.

Proof. Permutations act by parallel bundle isometries, which commute with ∇ and preserve Diff^{r-2}. Hence the set of σ for which the lemma holds contains the identity and is closed under composition with adjacent transpositions. It therefore suffices to treat an adjacent transposition $\tau_i = (i \; i{+}1)$ with $1 \leq i \leq r - 1$.

Write $\nabla^r u = \nabla^{i-1}(\nabla^2 S)$ with $S = \nabla^{r-i-1} u$, a section of $W' = \otimes^{r-i-1} T^*M \otimes W$. The Ricci identity states that $(\nabla^2 S)(X, Y) - (\nabla^2 S)(Y, X) = R^{W'}(X, Y) S$, where the curvature $R^{W'}$ of W' is expressed through Rm_g and R^W. The permutation τ_i exchanges exactly the two slots of $\nabla^2 S$, so

$$\tau_i \nabla^r u - \nabla^r u = \nabla^{i-1}(\mathcal{R} S),$$

where $\mathcal{R}$ is a zeroth-order bundle map. By the Leibniz rule, $\nabla^{i-1}(\mathcal{R} S) = \sum_{a+b=i-1} (\nabla^a \mathcal{R}) * \nabla^{b+r-i-1} u$, which is an operator of order $(i - 1) + (r - i - 1) = r - 2$. □

2.2. **Jets and nonlinear operators.**

Definition 2.2 (covariant jets)**.** For a Riemannian bundle W we set

$$\mathrm{J}^r W = \bigoplus_{k=0}^{r} \otimes^k T^*M \otimes W, \qquad j^r u = (u, \nabla u, \dots, \nabla^r u),$$

and call $j^r u$ the *covariant r-jet* of u. Elements of $\mathrm{J}^r W$ are written $\zeta = (\zeta_0, \dots, \zeta_r)$, and $\mathrm{J}^r W$ carries the direct-sum fibre metric.

Definition 2.3 (equation data)**.** The data of the equation are:

- an integer $p \geq 1$, a time $T_0 > 0$ and a Riemannian bundle V;
- an open subset $\mathcal{D} \subseteq [0, T_0] \times \mathrm{J}^{2p} V$ (open in the relative topology);
- a smooth map $F \colon \mathcal{D} \to V$ with $F(t, \zeta) \in V_x$ whenever $\zeta \in \mathrm{J}^{2p}_x V$.

For a time-dependent section u we say that u *has jets in* a set $\mathcal{S}$ if $(t, j^{2p}u(x,t)) \in \mathcal{S}$ for all (x,t). When u has jets in $\mathcal{D}$ we write $F[u](x,t) = F\big(t, j^{2p}u(x,t)\big)$.

Definition 2.4. Let $u_0 \in C^\infty(M; V)$ and $T \in (0, T_0]$. A *solution on $[0,T]$ with initial datum u_0* is a section $u \in C^\infty(M \times [0,T]; V)$ that has jets in $\mathcal{D}$ and satisfies $\partial_t u = F[u]$ and $u(\cdot, 0) = u_0$.

Definition 2.5 (linearization)**.** For $0 \leq k \leq 2p$ let $D_k F(t, \zeta) \in \mathrm{Hom}(\otimes^k T^*_x M \otimes V_x, V_x)$ denote the derivative of F along the summand $\otimes^k T^*M \otimes V$ of the fibre. For a smooth u with jets in $\mathcal{D}$ we define the linearization

$$L_u v = \sum_{k=0}^{2p} D_k F\big(t, j^{2p}u\big)(\nabla^k v), \qquad \text{so that} \qquad \frac{d}{d\varepsilon}\Big|_{\varepsilon=0} F[u + \varepsilon v] = L_u v, \tag{2.1}$$

and its principal coefficient $P_u = D_{2p} F(t, j^{2p}u)$.

2.3. **The Legendre condition.**

Definition 2.6 (Legendre condition)**.** Let W be a Riemannian bundle, $x \in M$ and $P \in \mathrm{Hom}(\otimes^{2p} T^*_x M \otimes W_x, W_x)$.

(1) Define $\widehat{P} \in \mathrm{End}(\otimes^p T^*_x M \otimes W_x)$ by

$$\langle \xi, \widehat{P}\eta \rangle = \sum_{I,J} \langle \xi_I, P(e^I \otimes e^J \otimes \eta_J) \rangle,$$

where I and J range over $\{1, \dots, n\}^p$ and (e_i) is an orthonormal basis of $T_x M$. The right-hand side does not depend on the basis.

(2) The *Legendre modulus* of P is

$$\kappa(P) = \min_{|\xi|=1} (-1)^{p+1} \langle \xi, \widehat{P}\xi \rangle.$$

(3) P satisfies the *Legendre condition* if $\kappa(P) > 0$.

The *parabolicity domain* of F is

$$\mathcal{O} = \{(t, \zeta) \in \mathcal{D} : \ \kappa\big(D_{2p} F(t, \zeta)\big) > 0\}.$$

A solution is *parabolic* if it has jets in $\mathcal{O}$.

The sign $(-1)^{p+1}$ is chosen so that $\partial_t u = -(\nabla^* \nabla)^p u$, the "heat equation of order $2p$", is parabolic; see Lemma 2.8(b).

Lemma 2.7. *(i) The map $P \mapsto \widehat{P}$ is linear. Moreover κ is superadditive and positively homogeneous: $\kappa(P + P') \geq \kappa(P) + \kappa(P')$ and $\kappa(cP) = c\,\kappa(P)$ for $c \geq 0$. In particular each set $\{\kappa \geq \lambda\}$ is convex.*

(ii) The set $\mathcal{O}$ is open in $[0,T_0]\times \mathrm{J}^{2p}V$.

(iii) Let $K\subseteq\mathcal{O}$ be compact. There are $\delta>0$ and $\kappa_K>0$ with the following property. Suppose $(t',\zeta')\in[0,T_0]\times\mathrm{J}^{2p}V$ and there is $(t,\zeta)\in K$ over the same point of M with $|t'-t|\le\delta$ and $|\zeta'-\zeta|\le\delta$. Then $(t',\zeta')\in\mathcal{O}$ and $\kappa(D_{2p}F(t',\zeta'))\ge\kappa_K$.

Proof. (i) The map $P\mapsto\widehat{P}$ is linear by definition, and a minimum of a sum is at least the sum of the minima.

(ii) The function $(t,\zeta)\mapsto\kappa(D_{2p}F(t,\zeta))$ is continuous on $\mathcal{D}$, because κ is continuous in P and in the base point, and $\mathcal{O}$ is where it is positive.

(iii) Let K_δ be the set of points (t',ζ') described in the statement. Then K_δ is compact, decreases with δ, and $\bigcap_\delta K_\delta = K$. The continuous function $\kappa\circ D_{2p}F$ is at least $2\kappa_K := \min_K \kappa\circ D_{2p}F>0$ on K, hence at least κ_K on an open neighbourhood $U\subseteq\mathcal{D}$ of K. Since the K_δ are compact, they are contained in U for δ small. □

Lemma 2.8 (examples)**.** *(a)* Product structure. *Let $E_1,\dots,E_p$ be symmetric forms on T^*_xM with $E_\ell\ge e_\ell\, g^{-1}$ for constants $e_\ell>0$. Let $B\in\mathrm{End}(W_x)$ be symmetric with $B\ge b>0$. Let*

$$P(X)=(-1)^{p+1}\sum E_1^{i_1j_1}\cdots E_p^{i_pj_p}\,B\,X_{i_1\dots i_pj_1\dots j_p}.$$

Then $\kappa(P)\ge b\,e_1\cdots e_p$.

(b) Polyharmonic operator. *There is $Q_0\in\mathrm{Diff}^{2p-2}(W,W)$ with*

$$-(\nabla^*\nabla)^p u = P_0(\nabla^{2p}u)+Q_0u,\qquad P_0(X)=(-1)^{p+1}\sum_I X_{II},$$

and $\kappa(P_0)=1$.

Proof. (a) Identify E_ℓ with a symmetric endomorphism of T^*_xM via g. Then $(-1)^{p+1}\widehat{P}=E_1\otimes\cdots\otimes E_p\otimes B$ on $\otimes^pT^*_xM\otimes W_x$. A tensor product of positive symmetric operators is positive, and its spectrum consists of products of eigenvalues of the factors.

(b) Since contraction commutes with ∇,

$$(\nabla^*\nabla)^pu=(-1)^p\sum_{i_1,\dots,i_p}(\nabla^{2p}u)_{i_1i_1i_2i_2\dots i_pi_p}.$$

Choose $\sigma\in S_{2p}$ that moves the slot pairs $(1,2),(3,4),\dots$ to the pairs $(1,p{+}1),(2,p{+}2),\dots$. Lemma 2.1 applied to σ yields $-(\nabla^*\nabla)^pu=(-1)^{p+1}\sum_I(\nabla^{2p}u)_{II}+Q_0u$. Finally, P_0 is the case $E_\ell=g^{-1}$, $B=\mathrm{id}$ of (a), and there $(-1)^{p+1}\widehat{P_0}=\mathrm{id}$. □

Remark 2.9 (rewriting the top-order term). The Legendre condition depends on how the top-order derivatives are arranged in F, not only on the equation. By Lemma 2.1 this dependence is harmless. Let P' be any smooth coefficient whose restriction to symmetric tensors agrees with that of $D_{2p}F$. Then $(P'-D_{2p}F)(\nabla^{2p}u)$ equals a lower-order expression in u, because $\nabla^{2p}u-\mathrm{Sym}\,\nabla^{2p}u=-\frac{1}{(2p)!}\sum_\sigma Q_\sigma u$. So F can be replaced by a function F' with $F'[u]=F[u]$ for every u and $D_{2p}F'=P'$.

Our theorems therefore apply whenever *some* such rearrangement satisfies the Legendre condition. For example, the operators of [25] are written with the pairing $A^{i_1j_1\cdots i_pj_p}\nabla^{2p}_{i_1j_1\cdots i_pj_p}$. After moving to the split ordering of Lemma 2.8(a) they satisfy the Legendre condition.

Remark 2.10 (Legendre condition and strong parabolicity). *Legendre implies strong parabolicity.* Taking $\xi=\eta^{\otimes p}\otimes v$ in Definition 2.6 shows that the Legendre condition implies strong (Petrovskii) parabolicity:

$$(-1)^{p+1}\langle v,P(\eta^{\otimes 2p}\otimes v)\rangle\ge\kappa(P)|\eta|^{2p}|v|^2.$$

Characterization for scalar equations. In the scalar case $W = \mathbb{R}$, and in view of Remark 2.9, an operator admits a Legendre rearrangement if and only if $(-1)^{p+1}\sigma_P(\eta) - \lambda|\eta|^{2p}$ is a sum of squares of forms of degree p in η for some $\lambda > 0$. Here $\sigma_P(\eta) = P(\eta^{\otimes 2p})$ is the symbol.

Consequences of Hilbert's theorem. By Hilbert's theorem [17], every nonnegative form is a sum of squares exactly for binary forms, quadratic forms, and ternary quartics. Hence for scalar equations with $p = 1$, or with $n \leq 2$, or with $n = 3$ and $p = 2$, strong parabolicity and the existence of a Legendre rearrangement are equivalent. In general they are not: a small perturbation of the Motzkin form [27] gives a strongly parabolic scalar operator of order six in dimension three with no Legendre rearrangement. Such operators are outside the scope of this paper. [double-check the precise form of this remark, in particular strict positivity versus SOS.]

2.4. **How to check the hypothesis in practice.** General statements about ellipticity are easy to get lost in, so here is what the Legendre condition amounts to and how to verify it for a given equation.

The underlying idea. Freeze the coefficients and insert a plane wave $u = e^{i\langle \xi, x\rangle}a(t)$. Each covariant derivative becomes a factor of ξ, and the top-order term acts on a like a matrix of size $|\xi|^{2p}$, the *symbol*. Parabolicity means that high-frequency waves are damped at a rate comparable to $|\xi|^{2p}$. For example:

- $\partial_t u = \Delta u$ has symbol $-|\xi|^2$, so high frequencies are damped;
- $\partial_t u = +\Delta^2 u$ has symbol $+|\xi|^4$, so high frequencies grow: this is a backward heat equation.

There are three standard ways to make this precise.

(1) *Strong parabolicity*: the symbol is uniformly negative. This only tests pure powers $\xi \otimes \cdots \otimes \xi$.
(2) *Gårding's inequality*: $-\langle \psi, L\psi\rangle \geq c\|\psi\|^2_{H^p} - C\|\psi\|^2$, that is, the operator dissipates energy up to lower-order errors. This is the only form in which ellipticity enters the proofs below: everything after Lemma 4.5 uses nothing else.
(3) *The Legendre condition*: the matrix $(-1)^{p+1}\widehat{P}$ is positive definite on *all* p-tensors, not only on pure powers.

Why Legendre is the convenient version. Integrating $\int\langle \psi, P(\nabla^{2p}\psi)\rangle$ by parts p times moves half of the derivatives onto the other factor. The result is $\int\langle \nabla^p\psi, \widehat{P}\,\nabla^p\psi\rangle$ up to lower-order terms (Lemma 4.4). Since $\nabla^p\psi$ can be an essentially arbitrary p-tensor, positivity of $\widehat{P}$ on all p-tensors is exactly what this integration by parts needs.

Deriving Gårding's inequality from strong parabolicity alone is also possible, but it requires Fourier analysis, freezing of coefficients and a partition of unity. In Fourier variables $\nabla^p\psi$ becomes a pure power, but only for constant coefficients. The difference between the two conditions is precisely the difference between "positive on pure powers" and "positive on everything".

A recipe.

(1) Linearize the equation and take the coefficient P of $\nabla^{2p}u$.
(2) Arrange its $2p$ indices so that the first p slots are paired with the last p slots, commuting covariant derivatives where necessary. This changes the equation only by lower-order terms (Remark 2.9).
(3) Check that the resulting matrix, multiplied by $(-1)^{p+1}$, is positive definite on V-valued p-tensors.

For the top-order terms occurring in geometric flows, namely $-(-\Delta)^p$ or a product $E_1\otimes\cdots\otimes E_p$ of positive definite forms, step (3) is automatic by Lemma 2.8.

Example 2.11 (the arrangement matters)**.** Consider $\partial_t u = -\Delta^2 u$ for scalar u, so $p = 2$ and $(-1)^{p+1} = -1$.

- *Adjacent pairing.* Writing $-\Delta^2 u = -g^{ab} g^{cd} \nabla_a \nabla_b \nabla_c \nabla_d u$ and splitting the slots as (a, b) and (c, d) gives the form $\xi \mapsto (\operatorname{tr}_g \xi)^2$ on 2-tensors. It vanishes on trace-free ξ, so the Legendre condition fails for this way of writing the operator.
- *Split pairing.* Exchanging the two middle derivatives costs only a curvature term of order two (Lemma 2.1). It gives $-\Delta^2 u = -g^{ac} g^{bd} \nabla_a \nabla_b \nabla_c \nabla_d u +$ (lower order). The form is now $\xi \mapsto g^{ac} g^{bd} \xi_{ab} \xi_{cd} = |\xi|^2$, so the Legendre condition holds with $\kappa = 1$.

This is Lemma 2.8(b) for $p = 2$.

If an equation is strongly parabolic but admits no Legendre arrangement (Remark 2.10), one can replace Lemma 4.5 by the general Gårding inequality. Nothing downstream changes.

2.5. **Main results.** Set

$$m_0 = \left\lfloor \frac{n}{2p} \right\rfloor + 4.$$

The spaces $H^k(M; V)$ and $\mathcal{P}^m(V)$ used in Theorem C are defined in Section 3.

Theorem A (existence and regularity)**.** *Let $u_0 \in C^\infty(M; V)$ with $(0, j^{2p} u_0(x)) \in \mathcal{O}$ for every $x \in M$. Then there are $T \in (0, T_0]$ and a parabolic solution $u \in C^\infty(M \times [0, T]; V)$ with initial datum u_0.*

Theorem B (uniqueness)**.** *Two parabolic solutions on $[0, T]$ with the same initial datum coincide.*

Theorem C (smooth dependence)**.** *Let u be a parabolic solution on $[0, T]$ with initial datum u_0, and let $m \geq m_0$. There are $\delta > 0$ and a C^∞ map*

$$\mathcal{S} \colon \{a \in H^{2pm}(M; V) : \|a - u_0\|_{H^{2pm}} < \delta\} \to \mathcal{P}^m(V)$$

with $\mathcal{S}(u_0) = u$ and the following property: for every smooth a in the domain, $\mathcal{S}(a)$ is a parabolic solution on $[0, T]$ with initial datum a.

Corollary 2.12 (continuous dependence, sequential form)**.** *Let u be a parabolic solution on $[0, T]$ with initial datum u_0. If $u_0^{(k)} \to u_0$ in $C^\infty(M; V)$, then for all large k the parabolic solution $u^{(k)}$ with initial datum $u_0^{(k)}$ exists on $[0, T]$, and $u^{(k)} \to u$ in $C^\infty(M \times [0, T]; V)$.*

The formal version of Corollary 2.12 states the convergence of every covariant derivative $\nabla^j(u^{(k)} - u)$, uniformly on $M \times [0, T]$. Convergence of mixed derivatives $\nabla^j \partial_t^i (u^{(k)} - u)$ then follows by induction on i from the equation, since the jets of u lie in a compact subset of $\mathcal{D}$. Theorem C itself, which involves the spaces H^{2pm} and $\mathcal{P}^m$, is not among the trusted formal statements (Section 8); it is used in the proof of Corollary 2.12. It is a goal of the formalization that has not yet been carried out; the Lean proof of Corollary 2.12 avoids it (see `blueprint/DECISIONS.md`).

Corollary 2.13 (maximal solutions)**.** *Under the hypothesis of Theorem A, let T^* be the supremum of the times $T \leq T_0$ for which a parabolic solution on $[0, T]$ exists. There is a unique section u on $M \times [0, T^*)$, or on $M \times [0, T_0]$ if $T^* = T_0$ is attained, whose restriction to every $[0, T]$ with $T < T^*$ is a parabolic solution.*

Proof. By Theorem B, parabolic solutions on different intervals agree where both are defined. □

Corollary 2.14 (Mantegazza–Martinazzi [25, Thm. 1.1])**.** *Let $V = M \times \mathbb{R}$ with its flat connection, and let*

$$F(t,\zeta) = A(t,\zeta_0,\dots,\zeta_{2p-1})(\zeta_{2p}) + b(t,\zeta_0,\dots,\zeta_{2p-1})$$

on $\mathcal{D} = [0,T_0] \times \mathrm{J}^{2p}V$. Suppose A has the product structure $A = (-1)^{p-1}E_1 \otimes \cdots \otimes E_p$ of [25], and each E_ℓ is locally uniformly positive definite, meaning that on every compact subset of its domain it is bounded below by a positive multiple of g^{-1}. After the rearrangement of Remark 2.9 we have $\mathcal{O} = \mathcal{D}$. In particular, every smooth initial datum has a unique smooth solution for short time, and it depends continuously on the initial datum in C^∞.

Proof. By Remark 2.9 (made exact by the Ricci identity, Lemma 2.1: permuting the $2p$ derivative slots changes $\nabla^{2p}u$ by a smooth function of the jet of order $2p-2$) we replace F by F' with the same value on jets of sections, whose principal coefficient is the split pairing $(-1)^{p+1}E_1 \otimes \cdots \otimes E_p$. By Lemma 2.8(a), with the sign $(-1)^{p-1} = (-1)^{p+1}$, every point of $\mathcal{D}$ lies in $\mathcal{O}$ for F'. Solutions of F and F' coincide, so Theorems A, B and Corollary 2.12 apply to F. In the formalization the E_ℓ are only assumed positive definite at each point; locally uniform positivity then follows by continuity. □

3. Function spaces

In this section W is a Riemannian bundle over M. Constants C may change from line to line. They depend only on the quantities indicated and on (M,g) and W.

3.1. Sobolev spaces of integer order.

Definition 3.1. Let $k \in \mathbb{N}$ and $u, f \in C^\infty(M;W)$.

(1) The positive-order norm is $\|u\|_{H^k}^2 = \sum_{i=0}^k \|\nabla^i u\|^2$. The space $H^k(M;W)$ is the completion of $C^\infty(M;W)$ in this norm; it embeds injectively in L^2 because ∇ is closable.

(2) The negative-order norm is

$$\|f\|_{H^{-k}} = \sup\{\langle f,\varphi\rangle : \varphi \in C^\infty(M;W),\ \|\varphi\|_{H^k} \le 1\}.$$

We only use these norms on smooth sections.

Lemma 3.2. *For smooth sections:*

(i) $\|u\|_{H^{k+1}}^2 = \|u\|^2 + \|\nabla u\|_{H^k}^2$ *for* $k \ge 0$.

(ii) $|\langle f,\varphi\rangle| \le \|f\|_{H^{-k}}\|\varphi\|_{H^k}$. *Moreover* $s \mapsto \|f\|_{H^s}$ *is nondecreasing on* $\mathbb{Z}$.

(iii) For every $s \in \mathbb{Z}$ *there is* C *with* $\|\nabla f\|_{H^{s-1}} \le C\|f\|_{H^s}$.

(iv) Let $A \in \mathrm{Diff}^r(W,W')$ *be a time-independent operator, or a single time slice of a time-dependent one. If* $s \ge 0$ *then* $\|Au\|_{H^s} \le C\|u\|_{H^{s+r}}$. *If* $r = 0$ *this also holds for* $s < 0$. *The constant depends on bounds for finitely many derivatives of the coefficients of* A.

Proof. (i) follows from the definition, since $\nabla^{i+1}u = \nabla^i(\nabla u)$. The inequality in (ii) is the definition, and monotonicity follows from $\|\varphi\|_{H^k} \le \|\varphi\|_{H^{k+1}}$.

(iii) For $s \ge 1$ this is (i). For $s = -k \le 0$, observe that $\langle \nabla f,\varphi\rangle = \langle f, \nabla^*\varphi\rangle$ and $\|\nabla^*\varphi\|_{H^k} \le \sqrt{n}\,\|\nabla\varphi\|_{H^k} \le \sqrt{n}\,\|\varphi\|_{H^{k+1}}$, because ∇^* is a contraction of ∇ and contractions commute with ∇.

(iv) For $s \ge 0$ this follows from the Leibniz rule. For $s < 0$ and $r = 0$ use duality: $\langle Au,\varphi\rangle = \langle u, A^\dagger\varphi\rangle$, where $A^\dagger$ is the pointwise adjoint, and apply the case $s \ge 0$ to $A^\dagger$. □

Lemma 3.3 (interpolation)**.** *For integers $0 < j < k$ and every $\varepsilon > 0$ there is $C = C(n,j,k,\varepsilon)$ such that*

$$\|\nabla^j u\| \le \varepsilon\|\nabla^k u\| + C\|u\| \qquad \text{for all smooth } u.$$

Consequently, for $k \geq 1$ and with $N_k(u) = \sum_{i \leq k} \|\nabla^i u\|$,

$$N_{k-1}(u) \leq \varepsilon \|\nabla^k u\| + C\|u\|, \qquad \|u\|^2_{H^{k-1}} \leq \varepsilon \|u\|^2_{H^k} + C\|u\|^2, \qquad \|u\|^2_{H^k} \leq 2\|\nabla^k u\|^2 + C\|u\|^2.$$

Proof. Let $a_i = \|\nabla^i u\|$. Integrating by parts,

$$a_i^2 = \langle \nabla(\nabla^{i-1}u), \nabla^i u\rangle = \langle \nabla^{i-1}u, \nabla^*\nabla^i u\rangle \leq \sqrt{n}\, a_{i-1}a_{i+1}.$$

If some a_i with $i \geq 1$ vanishes, then $a_j = 0$ for all $j \geq 1$ and there is nothing to prove. Otherwise $c_i = \log a_i + \frac{1}{4}(\log n)\, i^2$ is a convex sequence. Hence $a_j \leq n^{j(k-j)/4} a_0^{1-j/k} a_k^{j/k}$, and Young's inequality gives the claim. The consequences follow by summing over j, using $\|u\|^2_{H^{k-1}} \leq N_{k-1}(u)^2$ and $\|\nabla^k u\| \leq \|u\|_{H^k}$. □

Note that $\|u\|_{H^k} \leq N_k(u) \leq \sqrt{k+1}\, \|u\|_{H^k}$. We set $N_k = N_0$ for $k < 0$.

Lemma 3.4 (permuting derivatives in L^2)**.** *Let $r \in \mathbb{N}$ and let $\sigma \in S_r$ act on the r derivative slots of $\nabla^r u$. There is C with*

$$\|\sigma(\nabla^r u) - \nabla^r u\| \leq C\, N_{r-2}(u) \qquad \text{for all smooth } u.$$

Proof. Permutations of slots are fibrewise isometries. Hence the set of σ for which the estimate holds contains the identity and is closed under products, since $\|\sigma\sigma'\nabla^r u - \nabla^r u\| \leq \|\sigma'\nabla^r u - \nabla^r u\| + \|\sigma\nabla^r u - \nabla^r u\|$. It therefore suffices to treat an adjacent transposition τ_i. As in the proof of Lemma 2.1, $\tau_i\nabla^r u - \nabla^r u = \nabla^{i-1}(\mathcal{R}S)$ with $S = \nabla^{r-i-1}u$ and $\mathcal{R}$ of order zero: the Ricci identity handles the two slots created by $\nabla^2 S$, and ∇ commutes with permutations of the slots it does not create. By the Leibniz rule $|\nabla^{i-1}(\mathcal{R}S)| \leq C\sum_{b \leq i-1} |\nabla^b S|$ pointwise, and integrating gives the claim. □

Fact 3.5 (Sobolev embedding [3, 16])**.** *If $k > n/2 + i$, then $\sup_M |\nabla^i u| \leq C\|u\|_{H^k}$ for smooth u. Hence elements of H^k are C^i sections.*

Fact 3.6 (weak and strong derivatives)**.** *Let $Y_1, \dots, Y_q$ be smooth vector fields spanning T_xM at every x. Let $u \in L^2(M; W)$, and suppose that for every word $(a_1, \dots, a_d)$ with $d \leq k$ the weak derivative $\nabla_{Y_{a_1}} \cdots \nabla_{Y_{a_d}} u$ lies in L^2, defined by duality with the formal adjoint $\nabla^\dagger_Y \psi = -\nabla_Y \psi - (\operatorname{div} Y)\psi$. Then $u \in H^k(M; W)$ and*

$$\|u\|_{H^k} \leq C \sum_{d \leq k} \sum_{a_1, \dots, a_d} \|\nabla_{Y_{a_1}} \cdots \nabla_{Y_{a_d}} u\|.$$

The same holds for time-dependent sections with $L^2(0, T; \cdot)$ norms. This follows from Friedrichs mollification in charts.

3.2. **Parabolic Sobolev spaces.** Fix $T \in (0, T_0]$ and $m \in \mathbb{N}$.

Definition 3.7. For $u \in C^\infty(M \times [0, T]; W)$ let

$$\|u\|^2_{\mathcal{P}^m} = \sum_{\substack{j,i \geq 0 \\ 2pj + i \leq 2pm}} \int_0^T \|\partial_t^j \nabla^i u\|^2 dt = \sum_{j=0}^m \int_0^T \|\partial_t^j u\|^2_{H^{2p(m-j)}} dt.$$

- $\mathcal{P}^m(W)$ is the completion of $C^\infty(M \times [0, T]; W)$ in this norm. It embeds in $L^2(M \times [0, T]; W)$.
- $\mathcal{P}^m_0(W)$ is the closure in $\mathcal{P}^m(W)$ of $\{u \in C^\infty : u(\cdot, 0) = 0\}$.

Thus $\mathcal{P}^0 = L^2$, and $\mathcal{P}^m_0$ is a closed subspace of $\mathcal{P}^m$, so it is a Hilbert space.

Lemma 3.8. *(i) $\partial_t \colon \mathcal{P}^{m+1} \to \mathcal{P}^m$ and $\nabla^k \colon \mathcal{P}^m \to \mathcal{P}^{m-1}$ for $k \leq 2p$ are bounded. Hence $j^{2p} \colon \mathcal{P}^m(W) \to \mathcal{P}^{m-1}(\mathrm{J}^{2p}W)$ is bounded.*

(ii) $\mathcal{P}^{m+1} \hookrightarrow \mathcal{P}^m$ *with norm at most* 1.

(iii) Every $A \in \mathrm{Diff}^{2p}(W, W')$ *whose coefficients are smooth on* $M \times [0,T]$ *is bounded* $\mathcal{P}^m \to \mathcal{P}^{m-1}$. *In particular* $\partial_t - L$ *is bounded* $\mathcal{P}^m \to \mathcal{P}^{m-1}$ *for* $L \in \mathrm{Diff}^{2p}(W, W)$.

(iv) For $a \in H^{2pm}(M; W)$, *the constant extension* $(Ea)(x,t) = a(x)$ *satisfies* $\|Ea\|_{\mathcal{P}^m} = \sqrt{T}\,\|a\|_{H^{2pm}}$.

(v) If $m \geq 1$, *then* $u \mapsto u(\cdot, t)$ *extends to a bounded map* $\mathcal{P}^m \to L^2(M; W)$ *that is continuous in* t. *Moreover* $u(\cdot, 0) = 0$ *for* $u \in \mathcal{P}^m_0$.

Proof. (i) and (ii) are immediate from the definition: for instance $\partial_t^j \nabla^i \nabla^k u = \partial_t^j \nabla^{i+k} u$ and $2pj + i + k \leq 2pm$ when $2pj + i \leq 2p(m-1)$ and $k \leq 2p$. (iii) follows from the Leibniz rule in x and t. (iv) is a direct computation.

(v) For a Hilbert space X and $h \in C^1([0,T]; X)$ we have $\sup_t \|h(t)\|_X \leq C_T\big(\|h\|_{L^2 X} + \|h'\|_{L^2 X}\big)$. Apply this with $X = L^2$ and $h = u$. □

Lemma 3.9 (embedding)**.** *Let* $m \geq 1$ *and* $i, j \geq 0$ *with* $j \leq m-1$ *and* $2p(m-j-1) > n/2 + i$. *Then*

$$\sup_{M \times [0,T]} |\partial_t^j \nabla^i u| \leq C \|u\|_{\mathcal{P}^m} \qquad \text{for smooth } u.$$

Hence $\partial_t^j \nabla^i u$ *is continuous for every* $u \in \mathcal{P}^m$. *If smooth* u_k *form a Cauchy sequence in* $\mathcal{P}^m$ *for every* m, *then all* $\partial_t^j \nabla^i u_k$ *converge uniformly on* $M \times [0,T]$, *the limit* u *is in* $C^\infty(M \times [0,T]; W)$, *and* $\partial_t^j \nabla^i u = \lim_k \partial_t^j \nabla^i u_k$. *In particular, if* $u \in \mathcal{P}^m$ *for all* m, *then* $u \in C^\infty(M \times [0,T]; W)$.

Proof. Both $\partial_t^j u$ and $\partial_t^{j+1} u$ are bounded in $L^2(0,T; H^{2p(m-j-1)})$ by $\|u\|_{\mathcal{P}^m}$. Apply the inequality from the proof of Lemma 3.8(v) with $X = H^{2p(m-j-1)}$, and then Fact 3.5.

For the sequence, the estimate makes every $\partial_t^j \nabla^i u_k$ uniformly Cauchy. Fix a chart and a local frame $B_1, \dots, B_n$ of vector fields with constant coordinates. In the chart and the induced trivialization, the derivative of the reading of u_k along the ℓ-th coordinate is the reading of $\nabla_{B_\ell} u_k$ minus a Christoffel term that is linear in u_k with smooth coefficients, and its time derivative is the reading of $\partial_t u_k$. These converge uniformly, so the limit of the readings is C^1 with the limits as derivatives. The sequences $\partial_t u_k$ and $\nabla_{B_\ell} u_k$ are again Cauchy in every $\mathcal{P}^m$, so induction gives C^∞. Evaluating at the centre of the chart identifies $\partial_t u$ and $\nabla_{B_\ell} u$ with the limits of $\partial_t u_k$ and $\nabla_{B_\ell} u_k$. Since the B_ℓ span each tangent space, this identifies ∇u with $\lim_k \nabla u_k$, and by induction all $\partial_t^j \nabla^i u$.

An element $v \in \mathcal{P}^m$ is determined by its continuous representative: if $v_k \to v$ with v_k smooth and $v_k \to 0$ uniformly, then every $\partial_t^j \nabla^i v_k$ converges in L^2, and integrating by parts against test sections vanishing at $t = 0, T$ shows inductively that all these limits vanish, so $v = 0$.

For the last claim, note that the inclusions $\mathcal{P}^{m+1} \hookrightarrow \mathcal{P}^m$ are compatible with the embeddings into L^2. Apply the sequence statement to smooth approximations of u in $\mathcal{P}^k$, $k \to \infty$. □

Proposition 3.10 (algebra and composition)**.** *Let* $N \geq N_0 := \lfloor n/(2p) \rfloor + 3$.

(i) $\sup|u| \leq C\|u\|_{\mathcal{P}^N}$. *Moreover, for every smooth bilinear bundle map* $\beta \colon W_1 \times W_2 \to W_3$, *possibly depending on* t,

$$\|\beta(u,v)\|_{\mathcal{P}^N} \leq C \|u\|_{\mathcal{P}^N} \|v\|_{\mathcal{P}^N}.$$

(ii) Let E, E' *be Riemannian bundles,* $S \subseteq \mathbb{R}^d$ *open,* $\mathcal{U} \subseteq S \times [0,T] \times E$ *open, and* $\Phi \colon \mathcal{U} \to E'$ *smooth and fibre-preserving. Let* $\mathcal{V} = \{(s,u) \in S \times \mathcal{P}^N(E) : (s,t,u(x,t)) \in \mathcal{U} \text{ for all } (x,t)\}$, *which is open. Then the map*

$$\mathcal{V} \to \mathcal{P}^N(E'), \qquad (s,u) \mapsto \Phi_s(t,u)$$

is C^∞. *Its derivative in* u *is* $h \mapsto D_u \Phi_s(t,u)h$.

Proof. Step 1: a pointwise bound for low-order factors. Let $2pj + i \le pN$. Then $j \le N - 1$ and

$$2p(N - j - 1) - i = 2pN - 2p - (2pj + i) \ge pN - 2p > n/2$$

by the choice of N_0. By Lemma 3.9, $\sup|\partial_t^j \nabla^i u| \le C\|u\|_{\mathcal{P}^N}$.

Step 2: the product estimate (i). Expand $\partial_t^j \nabla^i \beta(u, v)$ with $2pj + i \le 2pN$ by the Leibniz rule. In each term the parabolic weights $2pj_1 + i_1$ and $2pj_2 + i_2$ of the two factors add up to at most $2pN$, so one of them is at most pN. Estimate that factor in L^∞ by Step 1 and the other in L^2.

Step 3: composition. Part (ii) is local, so we may work in charts and trivializations, with s as additional variables that are constant in (x, t). By the chain rule (Faà di Bruno), $\partial_{t,x}^\alpha[\Phi(t, x, u)]$ is a sum of terms

$$(\partial_{t,x}^\beta D_u^k \Phi)(t, x, u)\,[\partial^{\alpha_1} u, \dots, \partial^{\alpha_k} u], \qquad \beta + \alpha_1 + \cdots + \alpha_k = \alpha,\ |\alpha_i| \ge 1.$$

The coefficients are bounded, since u takes values in a compact subset of $\mathcal{U}$. At most one α_i has parabolic weight larger than pN, so the terms are estimated as in Step 2. This shows that $\Phi_*\colon u \mapsto \Phi(t, u)$ maps bounded subsets of $\mathcal{V}$ into bounded sets, first for smooth u and then, by continuity, for all u.

Step 4: differentiability. Taylor's formula

$$\Phi(u + h) - \Phi(u) - D\Phi(u)h = \int_0^1 (1 - \tau) D^2\Phi(u + \tau h)[h, h]\, d\tau,$$

combined with (i) and Step 3, gives a bound $C\|h\|^2_{\mathcal{P}^N}$ on bounded sets. So Φ_* is differentiable, and its derivative is multiplication by the $\mathcal{P}^N$-section $D\Phi(u)$. The map $u \mapsto D\Phi(u)$ is again a composition operator, so by induction Φ_* is C^∞. Joint smoothness in (s, u) follows in the same way. □

Corollary 3.11 (the nonlinear operator)**.** *Let $m \ge m_0 = N_0 + 1$. The set*

$$\mathcal{P}^m_{\mathcal{D}} = \{u \in \mathcal{P}^m(V) :\ (t, j^{2p}u(x, t)) \in \mathcal{D} \text{ for all } (x, t)\}$$

is open. The map $u \mapsto F[u]$ is C^∞ from $\mathcal{P}^m_{\mathcal{D}}$ to $\mathcal{P}^{m-1}(V)$, with derivative

$$v \mapsto \sum_k D_k F(t, j^{2p}u)(\nabla^k v).$$

The same holds for smooth families F_s depending on a parameter s, jointly in (s, u).

In the formalization the statement is local: near a smooth w with jets in $\mathcal{D}$, the map is defined on a ball of $\mathcal{P}^m$ around w, agrees with $u \mapsto F[u]$ on smooth u, is C^∞ at w (for each n it is C^n on a ball whose radius depends on n), and its derivative at w is $v \mapsto \sum_k D_k F(t, j^{2p}w)(\nabla^k v)$. This is what the implicit function theorem needs.

Proof. Compose the bounded linear map $j^{2p}\colon \mathcal{P}^m \to \mathcal{P}^{m-1}(\mathrm{J}^{2p}V)$ of Lemma 3.8(i) with Proposition 3.10, using $N = m - 1 \ge N_0$. In the formalization the derivatives $g_k(u) = D_\zeta^k F(t, j^{2p}u)[j^{2p}h_1, \dots, j^{2p}h_k]$ satisfy, for smooth u near w and smooth h_i, the bounds $\|g_k(u)(h)\| \le C\prod\|h_i\|$ and $\|g_k(u + h) - g_k(u) - g_{k+1}(u)(h, \cdot)\| \le C\|h\|^2$ (Taylor's formula with integral remainder at fibre level, and Proposition 3.10); a map with such Taylor bounds on a dense subspace extends to a C^n map on the completion. □

Lemma 3.12 (spatial composition)**.** *Let $k > n/2$. The map $a \mapsto F(t, j^{2p}a)$ is C^∞ from $H^{k+2p}(M; V)$ to $H^k(M; V)$, jointly in t. It is defined on the open set of those a whose jets lie in the slice $\mathcal{D}_t$.*

Proof. The proof of Proposition 3.10, without time derivatives. □

3.3. **Spectral facts.**

Fact 3.13 (eigenbasis [3])**.** *There is an L^2-orthonormal basis $(\varphi_i)_{i\ge1}$ of $L^2(M;W)$ consisting of smooth sections with $\nabla^*\nabla\varphi_i=\mu_i\varphi_i$, where $0\le\mu_1\le\mu_2\le\cdots\to\infty$.*

Fact 3.14 (elliptic estimate)**.** *For every $k\in\mathbb{N}$ there is C such that*

$$\|u\|_{H^{2k}}\le C\big(\|(\nabla^*\nabla)^k u\|+\|u\|\big)\qquad\text{for smooth } u.$$

This follows from Bochner-type integrations by parts together with Lemma 2.1, by induction on k.

Let Π_N denote the L^2-orthogonal projection onto the span of $\varphi_1,\dots,\varphi_N$. Since $\nabla^*\nabla$ is symmetric on smooth sections, $\Pi_N(\nabla^*\nabla)^k=(\nabla^*\nabla)^k\Pi_N$ on smooth sections.

4. Linear theory

4.1. **The class of operators and Gårding's inequality.**

Definition 4.1. Let W be a Riemannian bundle, $T\in(0,T_0]$, $\lambda>0$, $K\in\mathbb{N}$ and $\Lambda\ge0$. The class $\mathcal{L}(W;T,\lambda,K,\Lambda)$ consists of the operators

$$Lu=P(\nabla^{2p}u)+\sum_{k<2p}R_k(\nabla^k u)$$

with the following properties:

- the coefficients P and R_k are smooth on $M\times[0,T]$;
- $\kappa(P(x,t))\ge\lambda$ for all (x,t);
- the sum of $\sup|\partial_t^a\nabla^b P|$ and $\sup|\partial_t^a\nabla^b R_k|$ over $a+b\le K$ and all k is at most Λ.

Definition 4.2 (flip of a coefficient)**.** Let E,F be Riemannian bundles and $A\in\Gamma(\mathrm{Hom}(T^*M\otimes E,F))$, the first slot being the derivative slot. Its *flip* is $\check{A}\in\Gamma(\mathrm{Hom}(E,T^*M\otimes F))$,

$$\check{A}(w)(v,\cdot)=A(v^\flat\otimes w)(\cdot),\qquad v^\flat=g(v,\cdot).$$

Its *remainder* is $R_A=\mathrm{tr}_{12}\circ\nabla\check{A}\in\Gamma(\mathrm{Hom}(E,F))$, where tr_{12} contracts the two leading slots with g^{-1}.

Lemma 4.3 (one integration by parts)**.** *For $T\in C^\infty(M;F)$ and $W\in C^\infty(M;E)$,*

$$\langle T,A(\nabla W)\rangle=-\langle\nabla T,\check{A}\,W\rangle-\langle T,R_AW\rangle.$$

Proof. In an orthonormal frame, $\mathrm{tr}_{12}\big((\mathrm{id}_{T^*M}\otimes\check{A})D\big)=\sum_i A(e_i^\flat\otimes D(e_i,\cdot))=A(D)$ for every $D\in T^*M\otimes T^*M\otimes E$, because $\sum_i e_i^\flat\otimes D(e_i,\cdot)=D$. By the Leibniz rule $\nabla(\check{A}W)=(\nabla\check{A})W+(\mathrm{id}\otimes\check{A})(\nabla W)$, so $A(\nabla W)=\mathrm{tr}_{12}\,\nabla(\check{A}W)-R_AW$. Finally $\mathrm{tr}_{12}\,\nabla V=\operatorname{div}V$ and $\langle T,\operatorname{div}V\rangle=-\langle\nabla T,V\rangle$. □

Lemma 4.4 (integration by parts in the principal term)**.** *Let P be a smooth section of $\mathrm{Hom}(\otimes^{2p}T^*M\otimes W,W)$ and $\psi\in C^\infty(M;W)$. Then*

$$\langle\psi,P(\nabla^{2p}\psi)\rangle=(-1)^p\langle\nabla^p\psi,\widehat{P}\,\nabla^p\psi\rangle+\mathcal{E}(\psi),\qquad|\mathcal{E}(\psi)|\le C\,N_{p-1}(\psi)\,N_p(\psi),$$

where C depends on $\sup_{a\le p}|\nabla^a P|$ and on the curvature. If P depends smoothly on $t\in[0,T]$, then C can be chosen uniformly in t.

Proof. Apply Lemma 4.3 p times, each time moving the outermost derivative of $\nabla^{2p-j}\psi = \nabla(\nabla^{2p-j-1}\psi)$ onto $\nabla^j\psi$. With $P^{(0)} = P$ and $P^{(j+1)} = P^{\check{(j)}}$ this gives

$$\langle \psi, P(\nabla^{2p}\psi)\rangle = (-1)^p\langle \nabla^p\psi, P^{(p)}\nabla^p\psi\rangle - \sum_{j<p}(-1)^j\langle \nabla^j\psi, R_{P^{(j)}}\nabla^{2p-j-1}\psi\rangle.$$

Unwinding the definition, $P^{(p)}(\xi)(v_1,\dots,v_p,\cdot) = P(v_p^\flat\otimes\cdots\otimes v_1^\flat\otimes\xi)$. Expanding in an orthonormal frame over the first p slots and reindexing I by its reversal gives $\langle\rho\xi, P^{(p)}\eta\rangle = \langle\xi,\widehat{P}\eta\rangle$ pointwise, where ρ reverses the order of the first p slots. Hence

$$\langle\nabla^p\psi, P^{(p)}\nabla^p\psi\rangle = \langle\nabla^p\psi,\widehat{P}\,\nabla^p\psi\rangle + \langle\nabla^p\psi - \rho\nabla^p\psi, P^{(p)}\nabla^p\psi\rangle,$$

and the last term is at most $C\,N_{p-2}(\psi)\|\nabla^p\psi\|$ by Lemma 3.4. (For $p=1$, ρ is the identity.)

The error terms. Each $R_{P^{(j)}}$ has order zero and involves one derivative of P. In the term $\langle\nabla^j\psi, R\nabla^{2p-j-1}\psi\rangle$ move $p-j-1$ further derivatives from the right to the left, again by Lemma 4.3. This produces pairings of $\nabla^a\psi$ with $\nabla^b\psi$, $a\le p$, $b\le p-1$, whose coefficients involve at most p derivatives of P. Each is at most $C\,N_{p-1}(\psi)N_p(\psi)$. □

Lemma 4.5 (Gårding's inequality)**.** *Let $L\in\mathcal{L}(W;T,\lambda,K,\Lambda)$ with $K\ge p$. There is $C = C(n,p,\lambda,\Lambda,M,W)$ such that for all $t\in[0,T]$ and all $\psi\in C^\infty(M;W)$*

$$-\langle\psi, L(t)\psi\rangle \ge \frac{\lambda}{4}\|\psi\|_{H^p}^2 - C\|\psi\|^2.$$

Proof. Write $N_k = N_k(\psi)$ as in Lemma 3.3.

The principal term. By Lemma 4.4 and the Legendre condition $(-1)^{p+1}\langle\xi,\widehat{P}\xi\rangle\ge\lambda|\xi|^2$, applied pointwise to $\xi = \nabla^p\psi$,

$$-\langle\psi, P(\nabla^{2p}\psi)\rangle \ge \lambda\|\nabla^p\psi\|^2 - CN_{p-1}N_p.$$

Lower-order terms of order $k\le p$. Here $|\langle\psi, R_k\nabla^k\psi\rangle| \le \Lambda N_{p-1}N_p$.

Lower-order terms of order $p<k<2p$. Write $\nabla^k\psi = \nabla^{k-p}(\nabla^p\psi)$ and integrate by parts $k-p$ times. This produces terms $\langle\nabla^b R_k * \nabla^a\psi, \nabla^p\psi\rangle$ with $a+b = k-p\le p-1$, which are bounded by $CN_{p-1}N_p$.

Conclusion. So $\langle\psi, L\psi\rangle \le -\lambda\|\nabla^p\psi\|^2 + CN_{p-1}N_p$. Use $N_p\le N_{p-1}+\|\nabla^p\psi\|$ and $N_{p-1}\le\delta\|\nabla^p\psi\| + C_\delta\|\psi\|$ from Lemma 3.3 with δ small, and absorb: $\langle\psi,L\psi\rangle + \frac{\lambda}{2}\|\nabla^p\psi\|^2 \le C\|\psi\|^2$. Finally $\|\psi\|_{H^p}^2\le 2\|\nabla^p\psi\|^2 + C\|\psi\|^2$. □

4.2. **Energy estimates.** For $r\in\mathbb{N}$ let $W_r = \otimes^r T^*M\otimes V$, so that $W_0 = V$ and $W_{r+1} = T^*M\otimes W_r$.

Lemma 4.6 (differentiating an operator)**.** *Let $L\in\mathcal{L}(W_r;T,\lambda,K,\Lambda)$ with $K\ge 1$. There are an operator $\tilde{L}\in\mathcal{L}(W_{r+1};T,\lambda,K-1,\tilde{\Lambda})$ and a smooth $S_0\in\Gamma(\mathrm{Hom}(W_r,W_{r+1}))$, depending on t, with*

$$\nabla(Lu) = \tilde{L}(\nabla u) + S_0 u \qquad \text{for all } u.$$

The constant $\tilde{\Lambda}$ and the C^{K-1}-norm of S_0 are bounded in terms of Λ, r and the geometry.

Proof. The new principal coefficient. Let $P^{(1)}\colon \otimes^{2p}T^*M\otimes W_{r+1}\to W_{r+1}$ act by P on the $2p$ form slots and on the W_r factor, leaving the extra T^*M factor untouched. The first slot of $\nabla u\in\Gamma(T^*M\otimes W_r)$ is the derivative slot, and $\nabla^{2p}(\nabla u) = \nabla^{2p+1}u$.

Comparing the two expressions. By the Leibniz rule,

$$\nabla\big(P(\nabla^{2p}u)\big) = (\nabla P)(\nabla^{2p}u) + P_\bullet(\nabla^{2p+1}u),$$

where $P_\bullet$ applies P to slots $2,\dots,2p+1$. Moving the first slot to position $2p+1$ is a cyclic permutation γ, so by Lemma 2.1

$$P_\bullet(\nabla^{2p+1}u) = P^{(1)}\big(\nabla^{2p}(\nabla u)\big) + P^{(1)}(Q_\gamma u), \qquad Q_\gamma \in \mathrm{Diff}^{2p-1}.$$

Everything else, including $\nabla(R_k\nabla^k u)$, lies in $\mathrm{Diff}^{2p}(W_r, W_{r+1})$.

Rewriting in terms of ∇u. Every $A \in \mathrm{Diff}^{2p}(W_r, W_{r+1})$ can be written as $Au = A_0u + A'(\nabla u)$ with $A' \in \mathrm{Diff}^{2p-1}(W_{r+1}, W_{r+1})$, since $\nabla^k u = \nabla^{k-1}(\nabla u)$. Set $\tilde{L} = P^{(1)}\nabla^{2p} + A'$ and $S_0 = A_0$.

The Legendre condition. Decompose $\xi \in \otimes^p T^*M \otimes W_{r+1}$ as $\xi = \sum_c \xi^{(c)} \otimes e^c$ along the extra factor. Then $\langle \xi, \widehat{P^{(1)}}\xi\rangle = \sum_c \langle \xi^{(c)}, \widehat{P}\xi^{(c)}\rangle$, so $\kappa(P^{(1)}) \geq \kappa(P) \geq \lambda$. □

Proposition 4.7 (energy estimates)**.** *Let $K_k = p + k$. For every $k \in \mathbb{N}$ and $r \in \mathbb{N}$, every $L \in \mathcal{L}(W_r; T, \lambda, K_k, \Lambda)$, and every $u \in C^\infty(M \times [0,T]; W_r)$ with $u(\cdot, 0) = 0$, set $f = \partial_t u - Lu$. Then*

$$\max_{t\in[0,T]} \|u(t)\|^2_{H^k} + \int_0^T \|u\|^2_{H^{k+p}} dt \leq C \int_0^T \|f\|^2_{H^{k-p}} dt, \tag{4.1}$$

with $C = C(k, r, n, p, \lambda, \Lambda, T_0, M, V)$.

Proof. We argue by induction on k, for all r simultaneously.

Base case $k = 0$. By Lemma 4.5 and Lemma 3.2(ii),

$$\begin{aligned}\frac{d}{dt}\|u\|^2 &= 2\langle u, Lu\rangle + 2\langle u, f\rangle \leq -\frac{\lambda}{2}\|u\|^2_{H^p} + 2C\|u\|^2 + 2\|u\|_{H^p}\|f\|_{H^{-p}} \\ &\leq -\frac{\lambda}{4}\|u\|^2_{H^p} + 2C\|u\|^2 + \frac{4}{\lambda}\|f\|^2_{H^{-p}}.\end{aligned}$$

Since $u(0) = 0$, Gronwall's inequality gives $\max_t \|u(t)\|^2 \leq \frac{4}{\lambda} e^{2CT_0} \int_0^T \|f\|^2_{H^{-p}}$. Integrating the differential inequality over $[0,T]$ then bounds $\int_0^T \|u\|^2_{H^p}$.

Induction step from k to $k+1$. Let $L \in \mathcal{L}(W_r; T, \lambda, K_{k+1}, \Lambda)$ and $w = \nabla u$, so that $w(\cdot, 0) = 0$. By Lemma 4.6,

$$\partial_t w = \tilde{L}w + S_0 u + \nabla f, \qquad \tilde{L} \in \mathcal{L}(W_{r+1}; T, \lambda, K_k, \tilde{\Lambda}).$$

Applying the induction hypothesis to w, followed by Lemma 3.2(iii)–(iv), gives

$$\max_t \|w\|^2_{H^k} + \int_0^T \|w\|^2_{H^{k+p}} \leq C \int_0^T \Big(\|u\|^2_{H^{\max(k-p,0)}} + \|f\|^2_{H^{k+1-p}} \Big).$$

Next,

$$\int_0^T \|u\|^2_{H^{\max(k-p,0)}} \leq T_0 \max_t \|u\|^2_{H^k} \leq C \int_0^T \|f\|^2_{H^{k-p}} \leq C \int_0^T \|f\|^2_{H^{k+1-p}},$$

using the induction hypothesis for u (valid because $K_{k+1} \geq K_k$) and the monotonicity in Lemma 3.2(ii). Finally $\|u\|^2_{H^{k+1}} = \|u\|^2 + \|w\|^2_{H^k}$ and $\|u\|^2_{H^{k+1+p}} = \|u\|^2 + \|w\|^2_{H^{k+p}}$ by Lemma 3.2(i). Combined with the case $k = 0$, this gives (4.1) for $k+1$. □

Remark 4.8 (uniformity in the formalization). In the formalization the constant is not tracked through the class $\mathcal{L}$. Instead, Proposition 4.7 is proved, by the same induction, for all operators $L_\theta = (1-\theta)L_0 + \theta L_1$, $\theta \in [0,1]$, on the segment between two operators L_0, L_1 that are parabolic with the same constant λ, with one constant C for the whole segment. This is the uniformity used in the proof of Theorem 4.13. The base case needs Gårding's inequality uniformly in θ; it holds because $\langle u, L_\theta u\rangle$ is affine in θ. In the induction step, Lemma 4.6 is applied to L_0 and L_1 separately, and $\tilde{L}_\theta = (1-\theta)\tilde{L}_0 + \theta\tilde{L}_1$. The identity $\partial_t \nabla u = \nabla \partial_t u$ for smooth families is proved weakly, by pairing with test fields and integrating by parts. Proposition 4.10 is proved in the same form, uniformly along such segments.

Remark 4.9. Negative-order norms enter only through the forcing term. They let the induction start at $k = 0$ and reach the "maximal regularity" level $k = p$ by differentiation alone. No separate argument is needed for the second-order-in-time energy.

Proposition 4.10 (a priori estimate)**.** *Let $m \geq 1$ and $L \in \mathcal{L}(V; T, \lambda, K, \Lambda)$ with $K \geq (2p+1)m$. Then*

$$\|u\|_{\mathcal{P}^m} \leq C\|(\partial_t - L)u\|_{\mathcal{P}^{m-1}} \qquad \text{for all } u \in \mathcal{P}_0^m(V),$$

with $C = C(m, n, p, \lambda, \Lambda, T_0, M, V)$.

Proof. By density and Lemma 3.8(iii) it suffices to treat smooth u with $u(0) = 0$. Let $f = \partial_t u - Lu$.

Top spatial regularity. Proposition 4.7 with $k = 2pm - p$ gives

$$\int_0^T \|u\|_{H^{2pm}}^2 \leq C \int_0^T \|f\|_{H^{2p(m-1)}}^2 \leq C\|f\|_{\mathcal{P}^{m-1}}^2.$$

Time derivatives. For $1 \leq j \leq m$ we have

$$\partial_t^j u = \sum_{a=0}^{j-1} \binom{j-1}{a} (\partial_t^a L)(\partial_t^{j-1-a} u) + \partial_t^{j-1} f,$$

where $\partial_t^a L$ has the ∂_t^a-differentiated coefficients. Set $b = j - 1 - a \leq j - 1$. By Lemma 3.2(iv),

$$\|(\partial_t^a L)\partial_t^b u\|_{H^{2p(m-j)}} \leq C\|\partial_t^b u\|_{H^{2p(m-j)+2p}} \leq C\|\partial_t^b u\|_{H^{2p(m-b)}}.$$

Induction on j therefore bounds $\int_0^T \|\partial_t^j u\|_{H^{2p(m-j)}}^2$ by $C\|f\|_{\mathcal{P}^{m-1}}^2$. □

4.3. **The model operator.**

Proposition 4.11. *The operator $L_0 = -(\nabla^*\nabla)^p$ on V belongs to $\mathcal{L}(V; T, 1, K, \Lambda_0(K))$ for every K. For every $m \geq 1$, the map $\partial_t - L_0 \colon \mathcal{P}_0^m(V) \to \mathcal{P}^{m-1}(V)$ is surjective.*

Proof. Membership in the class. This is Lemma 2.8(b).

Solving mode by mode. Let $f \in C^\infty(M \times [0,T]; V)$ and $f_i(t) = \langle f(\cdot, t), \varphi_i \rangle$, with φ_i as in Fact 3.13. Set

$$c_i(t) = \int_0^t e^{-\mu_i^p (t-\tau)} f_i(\tau)\, d\tau.$$

Then c_i is smooth, $c_i(0) = 0$ and $c_i' = -\mu_i^p c_i + f_i$. Hence $u_N = \sum_{i \leq N} c_i \varphi_i$ is smooth, $u_N(0) = 0$, and $(\partial_t - L_0)u_N = \Pi_N f$.

Density. We claim that $\Pi_N f \to f$ in $\mathcal{P}^{m-1}$. Indeed $\partial_t^j \Pi_N f = \Pi_N \partial_t^j f$, and for a smooth section h of V, Fact 3.14 gives

$$\|h - \Pi_N h\|_{H^{2k}}^2 \leq C \sum_{i > N} (1 + \mu_i)^{2k} \langle h, \varphi_i \rangle^2.$$

The right-hand side tends to 0 and is bounded by $C\|h\|_{H^{2k}}^2$. Apply this with $h = \partial_t^j f(\cdot, t)$ and use dominated convergence in t. Since $(\partial_t - L_0)u_N = \Pi_N f$ with u_N smooth and $u_N(0) = 0$, and smooth sections are dense in $\mathcal{P}^{m-1}$, the range of $\partial_t - L_0 \colon \mathcal{P}_0^m \to \mathcal{P}^{m-1}$ is dense.

Surjectivity. By Proposition 4.10, $\|u\|_{\mathcal{P}^m} \leq C\|(\partial_t - L_0)u\|_{\mathcal{P}^{m-1}}$ on $\mathcal{P}_0^m$, which is complete, so the range is closed. Hence it is all of $\mathcal{P}^{m-1}$. □

4.4. **The method of continuity and the linear theorem.**

Lemma 4.12 (method of continuity [13, Thm. 5.2])**.** *Let X be a Banach space, Y a normed space and $T_0, T_1 \in \mathcal{B}(X,Y)$. Put $T_s = (1-s)T_0 + sT_1$, and suppose that $\|x\| \le C\|T_s x\|$ for all $s \in [0,1]$ and $x \in X$. If T_0 is surjective, then so is T_1.*

Proof. Suppose T_σ is surjective. By the hypothesis it is then invertible, with $\|T_\sigma^{-1}\| \le C$. For $|s-\sigma| < \big(C(\|T_0\| + \|T_1\|)\big)^{-1}$, the equation $T_s x = y$ is equivalent to the fixed-point problem

$$x = T_\sigma^{-1} y - (s-\sigma) T_\sigma^{-1}(T_1 - T_0)x,$$

whose right-hand side is a contraction. Hence T_s is surjective. Steps of this fixed length cover $[0,1]$. □

Theorem 4.13 (linear parabolic theory)**.** *Let $m \ge 1$ and $L \in \mathcal{L}(V; T, \lambda, K, \Lambda)$ with $K \ge (2p+1)m$.*

(i) The map $\partial_t - L \colon \mathcal{P}_0^m(V) \to \mathcal{P}^{m-1}(V)$ is a linear homeomorphism.

(ii) Suppose L has smooth coefficients and let $f \in C^\infty(M \times [0,T]; V)$ and $u_0 \in C^\infty(M; V)$. Then there is exactly one $u \in C^\infty(M \times [0,T]; V)$ with $\partial_t u = Lu + f$ and $u(0) = u_0$.

Proof. (i) Let $L_s = (1-s)L_0 + sL$. By Lemma 2.7(i), $L_s \in \mathcal{L}(V; T, \min(1,\lambda), K, \max(\Lambda_0, \Lambda))$ for all $s \in [0,1]$, so Proposition 4.10 holds uniformly in s. Lemma 4.12 and Proposition 4.11 give surjectivity. Injectivity and continuity of the inverse follow from Proposition 4.10.

(ii) Put $v = u - u_0$, with u_0 constant in time. We must find a smooth v with $v(0) = 0$ and $(\partial_t - L)v = h$, where $h = f + Lu_0$ is smooth.

- For each k, (i) at level $m = k+1$ gives $V \in \mathcal{P}_0^{k+1}$ with $(\partial_t - L)V = h$. Approximating V by smooth families vanishing at $t = 0$ gives a smooth v_k with $v_k(0) = 0$ and $\|(\partial_t - L)v_k - h\|_{\mathcal{P}^k}^2 < 1/(k+1)$.
- By Proposition 4.10 at level $q+1$, for $k, k' \ge q$, $\|v_k - v_{k'}\|_{\mathcal{P}^{q+1}}^2 \le C\|(\partial_t - L)(v_k - v_{k'})\|_{\mathcal{P}^q}^2 \le 2C\big(\frac{1}{k+1} + \frac{1}{k'+1}\big)$. So (v_k) is Cauchy in every $\mathcal{P}^q$.
- By Lemma 3.9, v_k converges with all derivatives $\partial_t^j \nabla^i$, uniformly, to a smooth v with $v(0) = 0$. Likewise $(\partial_t - L)v_k \to h$ uniformly.
- Passing to the limit in $\partial_t v_k = Lv_k + (\partial_t - L)v_k$ gives $\partial_t v = Lv + h$.

For uniqueness, the difference w of two smooth solutions is smooth, vanishes at $t = 0$ and solves $\partial_t w = Lw$. By Lemma 4.5, $\frac{d}{dt}\|w\|^2 = 2\langle w, Lw\rangle \le 2C\|w\|^2$, so $w = 0$ by Grönwall's inequality. (Alternatively, $w \in \mathcal{P}_0^m$ is zero by (i).) □

5. The nonlinear theory

Throughout this section F, $\mathcal{D}$ and $\mathcal{O}$ are as in Section 2, and $m \ge m_0$ is fixed.

5.1. **Approximate solutions.**

Lemma 5.1 (formal solution)**.** *Let $u_0 \in C^\infty(M; V)$ with $(0, j^{2p}u_0(x)) \in \mathcal{D}$ for all x. There are sections $a_0 = u_0, a_1, a_2, \ldots \in C^\infty(M; V)$ with the following property. Let $N \ge 1$, $T' > 0$, and let w be a smooth section on $M \times [0,T']$ with jets in $\mathcal{D}$ and $\partial_t^\ell w(0) = a_\ell$ for $\ell \le N$. Then*

$$\partial_t^\ell\big|_{t=0}\big(\partial_t w - F[w]\big) = 0 \qquad \text{for } 0 \le \ell \le N-1.$$

Proof. Since ∂_t commutes with ∇, we have $\partial_t^i j^{2p} w = j^{2p}\partial_t^i w$. By the chain rule there are smooth fibre maps Φ_ℓ, determined by F and its derivatives at $t = 0$, such that

$$\partial_t^\ell\big|_{t=0} F[w] = \Phi_\ell\big(j^{2p}a_0, \ldots, j^{2p}a_\ell\big)$$

for every such w: by the chain rule (Faà di Bruno), $\partial_t^\ell|_{t=0}F[w]$ depends only on $\partial_t^i w(0)$, $i \le \ell$. Define recursively $a_{\ell+1} = \partial_t^\ell|_{t=0}F[w_\ell]$ with $w_\ell = \sum_{i\le\ell} a_i t^i/i!$, taken on a short interval on which the jets of w_ℓ lie in $\mathcal{D}$; such an interval exists since $\mathcal{D}$ is open and M is compact, and one-sided derivatives at $t = 0$ do not depend on the interval. Then $\partial_t^\ell \partial_t w(0) = a_{\ell+1} = \partial_t^\ell F[w](0)$ whenever $\ell + 1 \le N$. □

Lemma 5.2 (cutoff)**.** *Let $f \in C^\infty(M \times [0,T]; V)$ with $\partial_t^\ell f(\cdot, 0) = 0$ for $\ell \le N-1$. Let $\chi \in C^\infty(\mathbb{R})$ with $\chi = 0$ on $(-\infty, 1]$ and $\chi = 1$ on $[2,\infty)$, and set $f_\varepsilon(x,t) = \chi(t/\varepsilon)f(x,t)$. Then f_ε is smooth, vanishes for $t \le \varepsilon$, and $f_\varepsilon \to f$ in $\mathcal{P}^{m'}$ as $\varepsilon \to 0$, for every $m' \le N$.*

Proof. A pointwise bound. By Taylor's theorem, $|\partial_t^l \nabla^i f(x,t)| \le C_i t^{N-l}$ for $l \le N$.

Expanding the difference. We have

$$\partial_t^j \nabla^i (f - f_\varepsilon) = \sum_{q\le j} \binom{j}{q} \partial_t^q \big(1 - \chi(t/\varepsilon)\big)\, \partial_t^{j-q} \nabla^i f,$$

which is supported in $t \le 2\varepsilon$.

- The terms with $q \ge 1$ are bounded by $C\varepsilon^{-q}(2\varepsilon)^{N-j+q} = C\varepsilon^{N-j}$, which is at most C for $j \le N$.
- The term with $q = 0$ is bounded.

Since the support has measure $O(\varepsilon)$, all L^2 norms are $O(\varepsilon^{1/2})$. □

5.2. **Existence.** Let u_0 satisfy the hypothesis of Theorem A. Take the sections a_ℓ of Lemma 5.1 and set

$$w(x,t) = \sum_{\ell\le m} a_\ell(x)\frac{t^\ell}{\ell!}.$$

Then $j^{2p}w(\cdot,0) = j^{2p}u_0$. By Lemma 2.7(iii) and continuity there are $T_1 \in (0, T_0]$ and $\lambda_1 > 0$ such that w has jets in $\mathcal{O}$ on $[0,T_1]$ and $\kappa(D_{2p}F(t, j^{2p}w)) \ge \lambda_1$ there. Let $\delta_1 > 0$ be such that the δ_1-tube (in the sense of Lemma 2.7(iii)) around the compact set $K_w = \{(t, j^{2p}w(x,t))\}$ is contained in $\mathcal{O}$.

From now on $T = T_1$ in all function spaces. Set $f = \partial_t w - F[w]$; it is smooth, and $\partial_t^\ell f(0) = 0$ for $\ell \le m-1$ by Lemma 5.1. Define

$$G(v) = \partial_t(w+v) - F[w+v] \qquad \text{on} \qquad \mathcal{U} = \{v \in \mathcal{P}_0^m :\ w + v \in \mathcal{P}_{\mathcal{D}}^m\}.$$

By Corollary 3.11, G is C^∞ from $\mathcal{U}$ to $\mathcal{P}^{m-1}$, with $G(0) = f$ and $DG(0) = \partial_t - L_w$. The linearization L_w has smooth coefficients and belongs to $\mathcal{L}(V; T, \lambda_1, K, \Lambda_K)$ for every K. By Theorem 4.13, $DG(0)$ is an isomorphism $\mathcal{P}_0^m \to \mathcal{P}^{m-1}$.

Proposition 5.3. *There are $\varepsilon > 0$ and $v \in \mathcal{P}_0^m$ with the following properties:*

- *$u = w + v$ has continuous jets lying in $\mathcal{O}$ for $t \le \varepsilon$;*
- *$\partial_t u = F[u]$ on $M \times [0,\varepsilon]$;*
- *$u(\cdot, 0) = u_0$.*

Moreover, $v = \Psi(0, f_\varepsilon)$, where Ψ is the map of Proposition 5.5 below, taken at the parameter value 0.

Proof. Inverting G. By the inverse function theorem (Fact A.1), G restricts to a C^∞ diffeomorphism from an open $\mathcal{U}' \ni 0$ onto an open $\mathcal{V}' \ni f$. Shrink $\mathcal{U}'$ so that $\sup|j^{2p}v| < \delta_1$ for $v \in \mathcal{U}'$; this is possible by Lemma 3.9, since $m - 1 \ge N_0$. By Lemma 5.2 with $N = m$ and $m' = m-1$, $f_\varepsilon \in \mathcal{V}'$ for ε small. Put $v = (G|_{\mathcal{U}'})^{-1}(f_\varepsilon)$.

The equation on $[0,\varepsilon]$. Since $f_\varepsilon = 0$ for $t \le \varepsilon$, we get $\partial_t u = F[u]$ almost everywhere on $M \times [0,\varepsilon]$. Both sides are continuous by Lemma 3.9, so equality holds everywhere.

Initial value and parabolicity. $u(0) = w(0) = u_0$ by Lemma 3.8(v). The jets of u lie within δ_1 of those of w, hence in $\mathcal{O}$. □

5.3. **Regularity by Angenent's trick.** The solution of Proposition 5.3 is only known to lie in $\mathcal{P}^m$. We now show that it is smooth. The idea is to transport it by flows of diffeomorphisms: the transported section solves a transported equation that depends smoothly on the flow parameter, and the implicit function theorem at the smooth point w turns this into smooth dependence on the parameter.

Horizontal flows. Let X be a smooth vector field on M with flow ϕ_s. Parallel transport along the curves $s \mapsto \phi_s(x)$ defines a one-parameter group $\tilde\phi_s$ of bundle isometries of V covering ϕ_s. The pull-back of a section is

$$(\tilde\phi_s^* u)(x) = \tilde\phi_s^{-1}\big(u(\phi_s(x))\big),$$

and it acts on time-dependent sections slice by slice. Given vector fields $X_1, \dots, X_d$, set $\tilde\phi_s = \tilde\phi_{s_1}^{X_1} \circ \cdots \circ \tilde\phi_{s_d}^{X_d}$ for $s \in \mathbb{R}^d$.

Lemma 5.4. *(a) For smooth u, the map $s \mapsto \tilde\phi_s^* u$ is C^∞ into $\mathcal{P}^{m'}$ for every m', and*

$$\partial_{s_1} \cdots \partial_{s_d}\big|_{s=0} \tilde\phi_s^* u = \nabla_{X_d} \cdots \nabla_{X_1} u.$$

(b) For $|s| \le 1$ the operators $\tilde\phi_s^*$ are uniformly bounded on $\mathcal{P}^{m'}$ and preserve $\mathcal{P}_0^{m'}$. For every $u \in \mathcal{P}^{m'}$ the map $s \mapsto \tilde\phi_s^* u$ is continuous into $\mathcal{P}^{m'}$.

(c) Suppose $u \in \mathcal{P}^{m'}$ and $s \mapsto \tilde\phi_s^* u$ is C^d into $\mathcal{P}^{m'}$ near 0. Then the weak derivative $\nabla_{X_d} \cdots \nabla_{X_1} u$ exists and equals $\partial_{s_1} \cdots \partial_{s_d}|_0 \tilde\phi_s^* u$, and hence lies in $\mathcal{P}^{m'}$.

(d) Naturality. *There are an open set $S \ni 0$ and smooth maps $F_s \colon \mathcal{D}_s \to V$ for $s \in S$, jointly smooth in (s, t, ζ) on an open set, with $F_0 = F$ and $\mathcal{D}_0 = \mathcal{D}$, such that*

$$\tilde\phi_s^*\big(F[u]\big) = F_s\big[\tilde\phi_s^* u\big]$$

for every smooth u with jets in $\mathcal{D}$. Here $F_s[\cdot]$ is formed with the jets of $\tilde\phi_s^* u$. Both sides are continuous in u from $\mathcal{P}_{\mathcal{D}}^m$ to $\mathcal{P}^{m-1}$, and smooth sections are dense in $\mathcal{P}_{\mathcal{D}}^m$, so the identity extends to all $u \in \mathcal{P}_{\mathcal{D}}^m$.

Proof. (a) The derivative of the pull-back by parallel transport along the flow lines of X is ∇_X. Joint smoothness is standard ODE theory.

(b) Use the chain rule and a change of variables, and approximate by smooth sections.

(c) Pair with a smooth test section ψ: $\langle \tilde\phi_s^* u, \psi\rangle = \langle u, (\tilde\phi_s^*)^\dagger \psi\rangle$, where $(\tilde\phi_s^*)^\dagger \psi$ is smooth in s. Differentiate at $s = 0$ using (a).

(d) Pulling back by $\tilde\phi_s$ intertwines ∇ with the pulled-back connections, and these differ from ∇ by tensors that depend smoothly on s and vanish at $s = 0$. Hence, in a chart, the jets of $\tilde\phi_s^* u$ are obtained from those of u by a recursion with smooth coefficients. Covariant jets of sections are not arbitrary (the antisymmetric part of $\nabla^2 u$ is a curvature term), so instead of inverting this jet map we use it in the other direction: there is a smooth family of fibrewise linear maps $\Upsilon_s(x) \colon \mathrm{J}^{2p}_x V \to \mathrm{J}^{2p}_{\phi_s(x)} V$, $\Upsilon_0 = \mathrm{id}$, with $\Upsilon_s(x)\big(j^{2p} v(x)\big) = j^{2p}(\tilde\phi_{-s}^* v)(\phi_s x)$ for all smooth v and $|s|$ small (glue the chart recursions with a partition of unity). Set $F_s(t, x, \zeta) = \tilde\phi_s(x)^{-1} F\big(t, \phi_s x, \Upsilon_s(x)\zeta\big)$, and let $\mathcal{D}_s$ be the corresponding preimage of $\mathcal{D}$. Naturality follows from $\tilde\phi_{-s}^* \tilde\phi_s^* = \mathrm{id}$. □

Proposition 5.5 (regularity)**.** *The solution u of Proposition 5.3 is smooth on $M \times [0, \varepsilon]$.*

Proof. Step 1: setting up the implicit function theorem. Fix smooth vector fields $Y_1, \dots, Y_q$ spanning TM at every point, a length $d \geq 1$ and a word $X_1, \dots, X_d$ in the Y's. Near $(0,0) \in \mathbb{R}^d \times \mathcal{P}_0^m$ consider

$$H(s,v) = \partial_t\big(\tilde\phi_s^* w + v\big) - F_s\big[\tilde\phi_s^* w + v\big] \in \mathcal{P}^{m-1}.$$

By Lemma 5.4(a),(d) and Corollary 3.11, H is C^∞. Moreover $H(0,\cdot) = G$ and $D_v H(0,0) = DG(0)$ is an isomorphism. By the implicit function theorem (Fact A.1) there are neighbourhoods $S' \times \mathcal{V}'$ of $(0,f)$ and $\mathcal{U}''$ of 0, and a C^∞ map $\Psi\colon S' \times \mathcal{V}' \to \mathcal{U}''$, such that for $(s,y) \in S' \times \mathcal{V}'$ and $v \in \mathcal{U}''$

$$H(s,v) = y \iff v = \Psi(s,y).$$

We choose ε in Proposition 5.3 so small that $f_\varepsilon \in \mathcal{V}'$, and take $v = \Psi(0, f_\varepsilon)$ there. This is consistent with the proof of that proposition, after shrinking $\mathcal{U}'$ to lie in $\mathcal{U}''$.

Step 2: smooth dependence on the flow parameter. By Lemma 5.4(d), $H(s, \tilde\phi_s^* v) = \tilde\phi_s^* f_\varepsilon$. By Lemma 5.4(a),(b), for s near 0 we have $\tilde\phi_s^* v \in \mathcal{U}''$ and $\tilde\phi_s^* f_\varepsilon \in \mathcal{V}'$. Therefore

$$\tilde\phi_s^* v = \Psi\big(s, \tilde\phi_s^* f_\varepsilon\big).$$

The right-hand side is C^∞ in s, because f_ε is smooth. By Lemma 5.4(c), $\nabla_{X_d} \cdots \nabla_{X_1} v \in \mathcal{P}^m$. Since the word was arbitrary and w is smooth, Fact 3.6 gives $\partial_t^j u \in L^2(0,\varepsilon; H^k)$ for all k and all $j \leq m$.

Step 3: regularity in time. By Lemma 3.8(v), applied with H^k in place of L^2, we get $u \in C([0,\varepsilon]; H^k)$ for all k. The equation $\partial_t u = F[u]$ says that

$$u(t) = u_0 + \int_0^t \mathfrak{F}\big(\tau, u(\tau)\big) d\tau \qquad \text{in } H^k,$$

where $\mathfrak{F}(t,a) = F(t, j^{2p}a)$ is a C^∞ map $H^{k+2p} \to H^k$, jointly in t, by Remark 3.12. Hence $u \in C^1([0,\varepsilon]; H^k)$ for all k. Inductively, $u \in C^j([0,\varepsilon]; H^k)$ for all j and k, and u is smooth by Fact 3.5. □

Proof of Theorem A. By Lemma 5.6 we may assume that $V = T_s^0 M$. Combine Propositions 5.3 and 5.5, with $T = \varepsilon$. □

5.4. **Uniqueness.**

Lemma 5.6 (lowering indices)**.** *Let $V = T_s^r M$. Lowering all indices with g is a parallel fibrewise isometry $\flat\colon T_s^r M \to T_{r+s}^0 M$. It carries equation data on V to equation data on $T_{r+s}^0 M$, parabolic solutions to parabolic solutions, and the Legendre condition to the Legendre condition with the same constant, and conversely through raising indices $\flat^{-1}$. In particular, Theorems A and B hold on every tensor bundle once they hold on every covariant tensor bundle $T_{s'}^0 M$.*

Proof. Since $\flat$ is parallel, $\nabla^k(\flat u) = \sigma_k \flat(\nabla^k u)$, where the slot permutation σ_k moves the k slots created by ∇^k in front of the lowered ones. Hence $\flat$ maps covariant jets to covariant jets by a fibrewise linear isometry $\flat_J$. Put $F^\flat(t,\zeta) = \flat F(t, \flat_J^{-1}\zeta)$ on $\mathcal{D}^\flat = \flat_J(\mathcal{D})$. Then u solves $\partial_t u = F[u]$ if and only if $\flat u$ solves $\partial_t(\flat u) = F^\flat[\flat u]$. Finally $D_{2p}F^\flat = \flat \circ D_{2p}F \circ (\sigma_{2p}\flat)^{-1}$. Prepending covectors commutes with $\sigma_k\flat$, and evaluating the first p slots of $\sigma_p \flat \xi$ gives $\flat$ of the same evaluation of ξ. Since $\flat$ is an isometry, the Legendre forms of $D_{2p}F$ at ξ and of $D_{2p}F^\flat$ at $\sigma_p\flat\xi$ agree. □

Lemma 5.7 (the difference of two solutions)**.** *Let $V = T_s^0 M$, let u and v be solutions on $[0,T]$, and let $[t_0, t_0+\eta] \subseteq [0,T]$. Suppose that for $t \in [t_0, t_0+\eta]$ and $\sigma \in [0,1]$ the jets $j^{2p}(v + \sigma(u-v))$*

lie in $\mathcal{D}$ and their principal coefficients satisfy the Legendre condition with constant λ. Then $h = u - v$ solves $\partial_t h = \tilde{L}h$ on $[t_0, t_0 + \eta]$, where

$$\tilde{L} = \sum_k \tilde{A}_k \nabla^k, \qquad \tilde{A}_k(t) = \int_0^1 D_k F\big(t, j^{2p}(v + \sigma h)\big) d\sigma,$$

and $\tilde{L}$ is a parabolic operator in the sense of Definition 4.1 with constant λ.

Proof. By the fundamental theorem of calculus along the segment of jets,

$$F(t, j^{2p}u) - F(t, j^{2p}v) = \sum_k \tilde{A}_k(\nabla^k h).$$

The Legendre form is linear and continuous in the coefficient, so the Legendre condition passes to the average $\tilde{A}_{2p}$.

The coefficients $\tilde{A}_k$ are smooth on $M \times [t_0, t_0 + \eta]$, up to the boundary. To see this, fix $x_0 \in M$ and work in the trivializations at x_0, over an open set $W \ni x_0$. There the components of $\tilde{A}_k(x,t)$ are $\int_0^1 f(x,t,\sigma)d\sigma$, where $f(x,t,\sigma)$ is the derivative, in the k-th jet slot, of the local representative of F at the coordinate jet of $v+\sigma h$. This f is smooth on $(W \times [t_0, t_0+\eta]) \times [0,1]$. Moreover, $q \mapsto \int_0^1 f(q,\sigma)d\sigma$ is smooth on $W \times S$, for S convex, whenever f is smooth on $(W \times S) \times [0,1]$: one-sided derivatives are differentiated under the integral sign. □

Lemma 5.8 (the Legendre form in coordinates)**.** *Let $V = T^0_s M$ and $x_0 \in M$. Read operators $P\colon (T^0_{s+2p})_x \to (T^0_s)_x$ and tensors $\xi \in (T^0_{s+p})_x$, for x near x_0, in the trivializations at x_0. Then, near x_0, the Legendre form $\langle \xi, \widehat{P}\xi\rangle$ and the norm $|\xi|^2$ are continuous functions of x and of the coordinates of P and ξ.*

Proof. The Legendre form is linear in P and bilinear in the tensor arguments. Using a bump function, choose smooth global sections $P_1, \dots, P_N$ and $\Psi_1, \dots, \Psi_K$ whose coordinates near x_0 are constant and form bases. In coordinates, $\langle \xi, \widehat{P}\xi\rangle$ is then a polynomial in the coordinates of P and ξ, with coefficients $\langle \Psi_j, \widehat{P}_i \Psi_l\rangle$. By the pointwise identity $\langle \rho\xi, P^{(p)}\eta\rangle = \langle \xi, \widehat{P}\eta\rangle$ from the proof of Lemma 4.4, each coefficient is the pointwise inner product of two smooth sections, hence continuous. The same argument applies to $|\xi|^2$. □

Lemma 5.9 (local uniqueness)**.** *Let $V = T^0_s M$, and let u and v be parabolic solutions on $[0,T]$ with $u(t^*) = v(t^*)$ for some $t^* < T$. Then there is $\eta > 0$ with $u = v$ on $[t^*, t^* + \eta] \cap [0,T]$.*

Proof. The segment between the jets stays parabolic. We show that there are $\eta > 0$ and $\lambda > 0$ such that, for $t \in [t^*, t^* + \eta]$, $x \in M$ and $\sigma \in [0,1]$, the jet $\zeta = j^{2p}(v + \sigma(u - v))(x,t)$ lies in $\mathcal{D}$ and $D_{2p}F(t,\zeta)$ satisfies the Legendre condition with constant λ. At $t = t^*$ the segment is the single jet $j^{2p}u(x,t^*)$, because $u(t^*) = v(t^*)$.

Fix $x_0 \in M$ and work in the trivializations at x_0. The jet of the segment is continuous in (x,t,σ), and at $t = t^*$ it lies in $\mathcal{D}$. Since $\mathcal{D}$ is relatively open and $[0,1]$ is compact, the segment stays in $\mathcal{D}$ for x near x_0 and t near t^*. There the coefficient $D_{2p}F$ along the segment is continuous in (x,t,σ). Let λ_0 be a Legendre constant of u at (t^*, x_0), and set

$$G(x,t,\sigma,\xi) = (-1)^{p+1}\langle \xi, \widehat{P}\xi\rangle - \tfrac{\lambda_0}{2}|\xi|^2, \qquad P = D_{2p}F\big(t, j^{2p}(v + \sigma(u-v))(x,t)\big),$$

with ξ in coordinates. By Lemma 5.8, G is continuous. At $x = x_0$ and $t = t^*$ it is positive for $\sigma \in [0,1]$ and $|\xi| = 1$. Since $[0,1]$ times the unit sphere is compact, $G > 0$ there for x near x_0 and $t \in [t^*, t^* + \eta_{x_0}]$. Both terms of G are quadratic in ξ, so the Legendre condition holds there with constant $\lambda_0/2$. Finitely many of these neighbourhoods cover M; take the least η_{x_0} and the least constant.

The difference solves a linear equation. Let $h = u - v$. Then $\partial_t h = \tilde{L}h = \sum_k \tilde{A}_k(\nabla^k h)$, where

$$\tilde{A}_k = \int_0^1 D_k F\big(t, j^{2p}(v + \sigma h)\big)\, d\sigma$$

is smooth, and $\tilde{L}$ is parabolic with constant λ on $[t^*, t^* + \eta]$, by Lemma 5.7.

Energy estimate. By Lemma 4.5, $\frac{d}{dt}\|h\|^2 = 2\langle h, \tilde{L}h\rangle \le 2C\|h\|^2$. Since $h(t^*) = 0$, Gronwall's inequality gives $h = 0$ on $[t^*, t^* + \eta]$. □

Proof of Theorem B. By Lemma 5.6 we may assume that $V = T^0_s M$. Let u and v be parabolic solutions on $[0, T]$ with the same initial datum. The set of $t \in [0, T]$ with $u(t) = v(t)$ is closed, because u and v are continuous in t, and it contains 0. By Lemma 5.9, each of its points $t^* < T$ is followed by an interval $[t^*, t^* + \eta]$ inside the set. Since $[0, T]$ is connected, the set is all of $[0, T]$. □

5.5. **Smooth dependence.**

Proof of Theorem C. Step 1: the implicit function theorem around u. Let u be a parabolic solution on $[0, T]$, and set $\bar{u} = u - Eu_0$. It is smooth with $\bar{u}(0) = 0$, so $\bar{u} \in \mathcal{P}^m_0$. On a neighbourhood of $(u_0, 0)$ in $H^{2pm}(M; V) \times \mathcal{P}^m_0$ define

$$H(a, v) = \partial_t(Ea + \bar{u} + v) - F[Ea + \bar{u} + v].$$

This is well defined and C^∞ by Lemma 3.8(iv) and Corollary 3.11. We have $H(u_0, 0) = 0$, and $D_v H(u_0, 0) = \partial_t - L_u$ is an isomorphism by Theorem 4.13, applied with the constant κ_K of the compact set of jets of u. The implicit function theorem gives $\delta > 0$ and a C^∞ map $a \mapsto v(a)$ with $H(a, v(a)) = 0$. Put

$$\mathcal{S}(a) = Ea + \bar{u} + v(a).$$

Then $\mathcal{S}(a)$ solves the equation on $[0, T]$ with $\mathcal{S}(a)(0) = a$. Its jets are uniformly close to those of u, hence in $\mathcal{O}$, when δ is small.

Step 2: smoothness for smooth data. Let a be smooth. Repeat the proof of Proposition 5.5 with the parameters (a, s) and

$$H(a, s, v) = \partial_t\big(Ea + \tilde{\phi}^*_s \bar{u} + v\big) - F_s\big[Ea + \tilde{\phi}^*_s \bar{u} + v\big].$$

Using $\tilde{\phi}^*_s E = E\tilde{\phi}^*_s$ one gets $\tilde{\phi}^*_s v(a) = \Psi(\tilde{\phi}^*_s a, s)$ for small s. This is C^∞ in s, because $s \mapsto \tilde{\phi}^*_s a$ is C^∞ into H^{2pm}. Hence $\mathcal{S}(a)$ is smooth, arguing as in Steps 2–3 of that proof on all of $[0, T]$. □

Proof of Corollary 2.12. Remark on the formalization. The Lean proof does not use the smooth map $\mathcal{S}$ of Theorem C. It perturbs the fixed smooth solution u by the time-independent family $a - u_0$, applies the implicit function theorem with parameter, and uses regularity in trace form at the fixed base solution (see `blueprint/DECISIONS.md`). The argument below is the original route.

- Consistency across levels: for $m' \ge m$ and smooth a in both domains, the level-m' and level-m solutions coincide by Theorem B.
- If $u_0^{(k)} \to u_0$ in C^∞, then for every m we have $\|u_0^{(k)} - u_0\|_{H^{2pm}} < \delta_m$ for large k.
- Therefore $u^{(k)} = \mathcal{S}_m(u_0^{(k)}) \to u$ in $\mathcal{P}^m$ for every m, and hence in C^∞ by Lemma 3.9.

□

6. Examples

In each example we identify the principal coefficient and check the Legendre condition using Lemma 2.8. The geometric conclusions that go beyond short-time existence are not discussed; an example is the invariance of a target manifold, or the passage from a gauge-fixed flow to the original flow.

6.1. **Quasilinear equations of product type.** This is the setting of Corollary 2.14. It includes the Q-curvature flow on four-manifolds, whose principal part is $-e^{-4w}\Delta_{g_0}^2$ (see [5, 24]), and the Willmore-type flows of [19, 28], once these are written as scalar equations for a graph function. [Check the exact forms against the references.]

6.2. **Polyharmonic heat flows of maps.** Let $N \subset \mathbb{R}^K$ be a closed submanifold, and let $V = M \times \mathbb{R}^K$ be the trivial bundle with the flat connection. The extrinsic polyharmonic map heat flow of order $2p$ has the form

$$\partial_t u = -(\nabla^*\nabla)^p u + \mathcal{Q}\big(u, \nabla u, \dots, \nabla^{2p-1}u\big),$$

with $\mathcal{Q}$ smooth for u in a tubular neighbourhood of N; see [10] for $p = 1$ and [20, 12] for $p \geq 2$. [Verify that the lower-order structure is as stated for all p.] By Lemma 2.8(b), $\mathcal{O} = \mathcal{D}$ is the set of jets whose zeroth component lies in the tubular neighbourhood. Theorems A–C give short-time existence, uniqueness and smooth dependence for every p, including the sixth-order ($p = 3$) triharmonic flow.

6.3. **The Ricci–DeTurck flow.** Fix a background metric $\tilde{g}$ on M, and use it as the metric g of Section 2. Let $V = \operatorname{Sym}^2 T^*M$ with the induced metric and connection. The Ricci–DeTurck flow [8, 15] can be written as

$$\partial_t g_{ij} = (g^{-1})^{ab}\tilde{\nabla}_a\tilde{\nabla}_b g_{ij} + \mathcal{Q}(g, g^{-1}, \tilde{\nabla}g, \widetilde{\mathrm{Rm}})_{ij},$$

with $\mathcal{Q}$ smooth; see [30, 7]. Here $\mathcal{D} = \mathcal{O}$ is the set of jets whose zeroth component is positive definite. The principal coefficient is $P(X) = (\zeta_0^{-1})^{ab}X_{ab}$, and $\kappa(P)$ is the smallest eigenvalue of ζ_0^{-1} relative to $\tilde{g}$, by Lemma 2.8(a) with $p = 1$. Theorems A–C therefore give short-time existence, uniqueness and smooth dependence for the Ricci–DeTurck flow. The Ricci flow is recovered by DeTurck's diffeomorphisms, which is a separate ODE argument.

6.4. **Parabolic Monge–Ampère equations.** A fully nonlinear example with a nontrivial parabolicity domain is

$$\partial_t u = \log \frac{\det\big(g + \nabla^2 u\big)}{\det g} + h(x, t, u, \nabla u),$$

for scalar u. Take $\mathcal{D} = \mathcal{O}$ to be the set of jets with $g + \operatorname{Sym}\zeta_2 > 0$. The principal coefficient is $P(X) = \big((g + \operatorname{Sym}\zeta_2)^{-1}\big)^{ab}X_{ab}$, which satisfies the Legendre condition on $\mathcal{D}$.

The Kähler–Ricci flow written as a parabolic complex Monge–Ampère equation for a potential [6] has the same structure. On a Kähler manifold, $\sqrt{-1}\partial\bar{\partial}\varphi$ is a linear function of $\nabla^2\varphi$, and the principal coefficient is a positive multiple of g_φ^{-1} acting on the J-invariant part. [Write out the Kähler case in detail.]

7. Remarks on the scope

(1) *Gauge-degenerate flows.* For the Ricci flow, the mean curvature and Willmore flows of immersions, the Yang–Mills flow and similar equations, F is degenerate in the directions of a gauge group, so the Legendre condition fails. The standard remedy (DeTurck's trick) is to add a gauge-fixing term, solve the resulting strictly parabolic equation, which is covered by our results, and then undo the gauge by solving an ODE. A general version of this last step, stated abstractly for flows equivariant under a gauge group, would be a natural companion to the present results.

(2) *Strong parabolicity without a Legendre representation.* By Remark 2.10, such operators exist in general. They require Gårding's inequality in its general form (localization and Fourier analysis). The rest of the argument is unchanged, because everything downstream of Lemma 4.5 uses only Gårding's inequality, and uses it only for operators $L^{(r)}$ obtained by differentiating (Lemma 4.6).

(3) *Nonlocal lower-order terms.* Suppose F depends in addition on finitely many integrals of the form $\int_M \Phi(t, j^{2p-1}u)\, d\mu$, such as normalizations or averages. Then Proposition 3.10 and the rest of the argument go through with trivial changes.

(4) *Analytic dependence.* If F is real analytic in ζ, the maps in the implicit function theorem are analytic, and $\mathcal{S}$ in Theorem C is analytic.

(5) *General bundles versus tensor bundles.* The paper is written for arbitrary Riemannian bundles. The formal statements treat the tensor bundles $V = T^r_s M$ of type (r, s), with the metric and connection induced by g and its Levi-Civita connection. These are closed under the operation $W \mapsto T^*M \otimes W$ used in Lemma 4.6, and for them the horizontal flows of Section 5.3 can be replaced by natural pull-backs of tensors. Vector-valued systems, $V = T^r_s M \otimes \mathbb{R}^K$ with $K \geq 2$, are not covered by the formal statements; they need only an extra fixed factor in the model fibre.

8. Notes on the formalization

8.1. **Dependency structure.** Every numbered item of this paper carries two annotations for software, invisible in the printed version: the Lean declarations formalizing it, and the items its statement and proof use. A script in the repository turns them into a dependency graph with formalization status, which it writes to `blueprint/map.md`. It can also *pin* approved Lean statements, so that later changes to them are detected. At the coarsest level the proof has four layers, each of which uses only the ones before it.

L1 *Geometry and algebra:* Lemma 2.1, Definition 2.6, Lemmas 2.7 and 2.8.

L2 *Function spaces:* Lemmas 3.2, 3.3, 3.8 and 3.9, Proposition 3.10, Corollary 3.11, and the imported Facts 3.5, 3.6, 3.13 and 3.14.

L3 *Linear theory:* Lemmas 4.4, 4.5, 4.6 and 4.12, Propositions 4.7, 4.10 and 4.11, and Theorem 4.13.

L4 *Nonlinear theory:* Lemmas 5.1, 5.2 and 5.4, Propositions 5.3 and 5.5, Theorems A, B and C, and the imported Fact A.1.

Layers L3 and L4 interact only through Theorem 4.13, and only for operators with smooth coefficients. The two layers can be developed in parallel once that statement is fixed.

8.2. **The trusted statements.** Theorems A and B, and Corollary 2.12 (the sequential form of Theorem C), mention only smooth sections, covariant derivatives, time derivatives and the pointwise algebraic Definition 2.6. These are the statements that should appear in the final formal version, so that the definitions a reader has to trust stay small. The parabolic Sobolev spaces are internal to the proofs.

8.3. **Imported facts and candidate sources.** The following table records where each imported or reusable ingredient might come from. The entries marked "to check" are based on file names and statements seen in the library, not on a build.

Ingredient	Used in	Candidate source
Ricci identity, commutators	Lemma 2.1	[9] `Tensor/RicciIdentity` (partial)
Divergence theorem, integration by parts	throughout	[9] `Analysis/Integration`
Iterated covariant derivatives	§2	[9] `iteratedCovGrad`
Interpolation (log-convexity)	Lemma 3.3	[9] `Sobolev/GagliardoNirenberg/FiberNorm` (to check)
Sobolev embedding	Fact 3.5	[9] `Sobolev/Embedding/Tensor`
Weak = strong derivatives	Fact 3.6	[9] `Sobolev/Intrinsic` (to check)
Eigenbasis of $\nabla^*\nabla$	Fact 3.13	[9] `Spectral/Tensor` (tensor bundles)
Elliptic estimate	Fact 3.14	[9] `Spectral/Tensor/SobolevScale/Jet` (to check)
$H^1(0,T;X) \subset C([0,T];X)$	Lemma 3.8	[9] `Parabolic/TimeSobolev/H1`
Per-mode ODE solution	Proposition 4.11	[9] `Parabolic/MaximalRegularity/PerMode`
Flows of vector fields	Lemma 5.4	[9] `Analysis/ODE`, Mathlib
Inverse/implicit function theorem	Fact A.1	Mathlib
Gronwall's inequality	Prop. 4.7, Thm. B	Mathlib
Method of continuity	Lemma 4.12	new (short)

8.4. **Points in this draft that need checking.**

- Remark 2.10: the precise statement and the references.
- Section 6: the exact form of each example equation against the literature.
- Lemma 5.4 and Step 3 of Proposition 5.5: these are written at the level of "routine", and a formalization will need them spelled out.
- The values of the constants K required in Propositions 4.7 and 4.10 are generous rather than sharp; any finite value suffices.

Appendix A. Abstract tools

Fact A.1 (implicit function theorem [21])**.** *Let X, Y and Z be Banach spaces, $U \subseteq Z \times X$ open, and $H\colon U \to Y$ of class C^k with $k \geq 1$. Suppose $H(z_0, x_0) = y_0$ and $D_x H(z_0, x_0)\colon X \to Y$ is invertible. Then there are open neighbourhoods $Z' \ni z_0$, $Y' \ni y_0$ and $X' \ni x_0$, and a C^k map $\Psi\colon Z' \times Y' \to X'$, such that for $(z, y, x) \in Z' \times Y' \times X'$*

$$H(z, x) = y \iff x = \Psi(z, y).$$

Taking Z to be a point gives the inverse function theorem.

Fact A.2 (Gronwall)**.** *Let $\varphi \in C^1([t_0, t_1])$ with $\varphi(t_0) = 0$ and $\varphi' \leq a\varphi + b$, where $a \geq 0$ and $b \geq 0$ is continuous. Then $\varphi(t) \leq e^{a(t-t_0)} \int_{t_0}^{t} b$.*

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